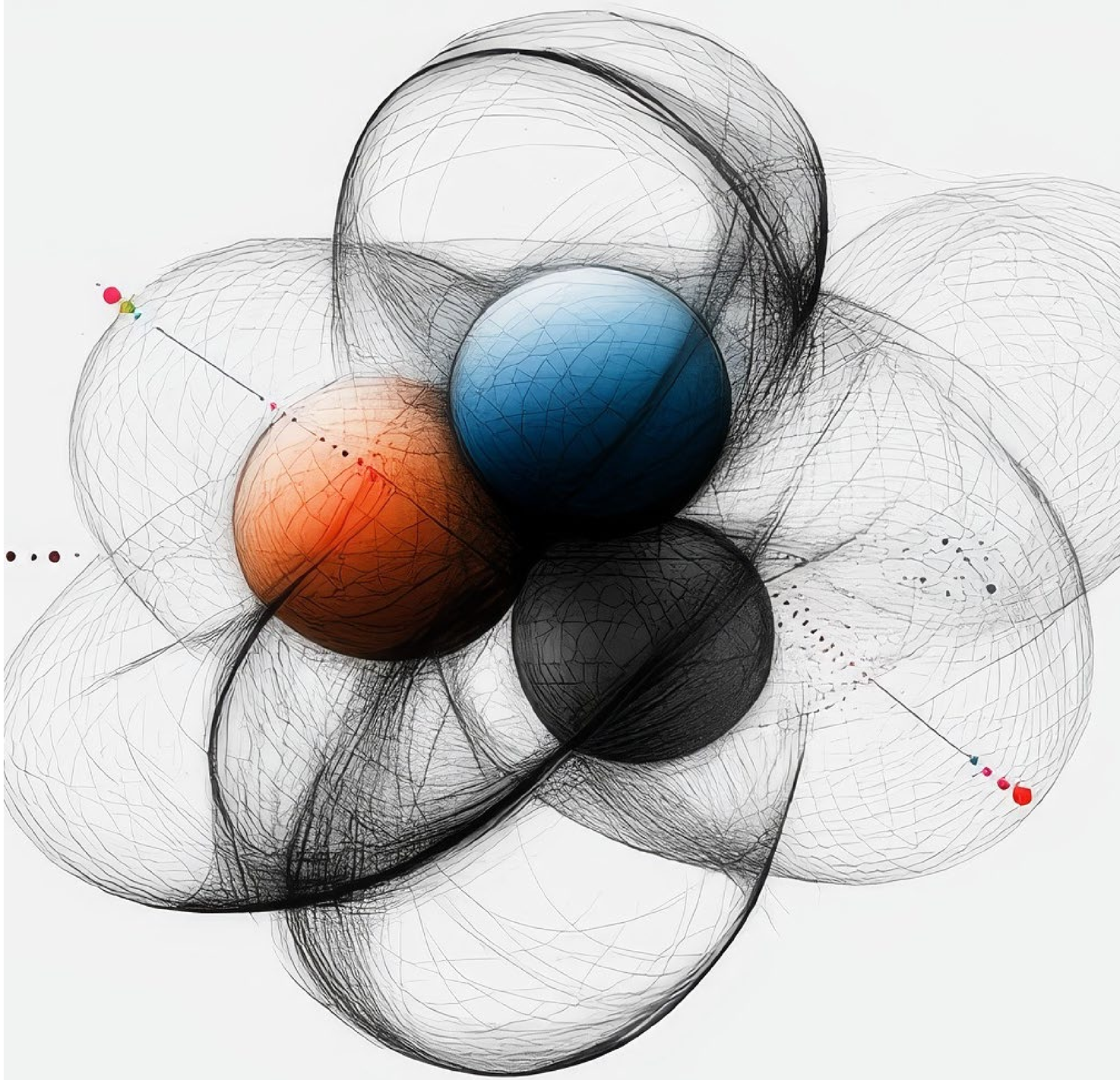
HSLU Lucerne University of Applied Sciences and Arts

Self Study: Atomic Models (Self Study)

Dr. Macarena San Martin Ruiz

HS25



Engineering and Architecture

September 29, 2026

FH Zentralschweiz

Review: What is an atom?

Where does the term “atom” come from?

What parts is it made of?

How heavy is an atom?

Size and mass relationships:

The atomic nucleus has a diameter of about one hundred-thousandth of the total atomic diameter.

The mass of the electrons is about 1/2000 that of a proton. Therefore, electrons have practically no effect on the mass of an atom

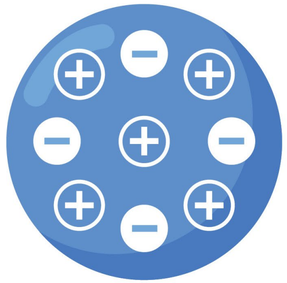
The Atomic Models

DALTON
1808



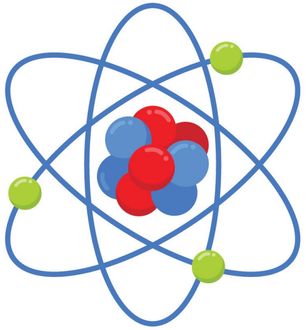
All matter is made of atoms

THOMSON
1897



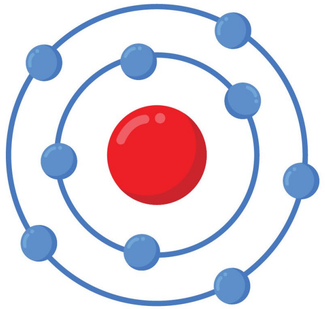
Discovered the inner structure of the atom

RUTHERFORD
1911



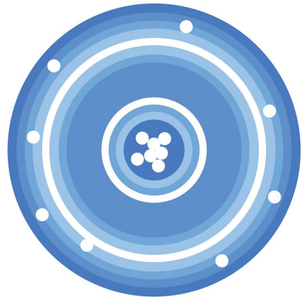
Discovered electrons surround the nucleus of an atom

BOHR
1913



Electron move is fixed orbitals (shells)

SCHRODINGER
1926

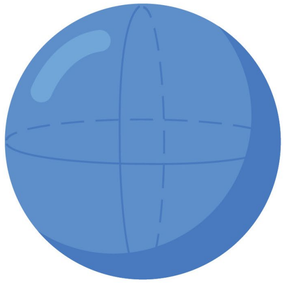


Wave behavior of the electron

First atomic model by Dalton

- Every substance consists of atoms.
- Atoms of an element have the same mass and size.
- Atoms cannot be newly created or destroyed. In chemical reactions, atoms are only rearranged.

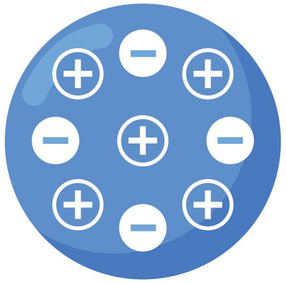
DALTON
1808



All matter is made of atoms

Development of the atomic model by Thomson

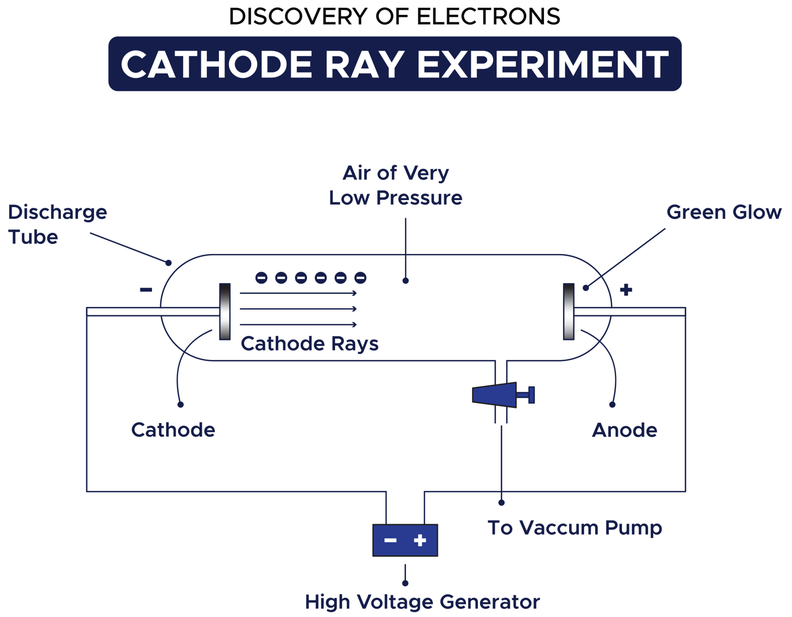
THOMSON
1897



Discovered the inner structure of the atom

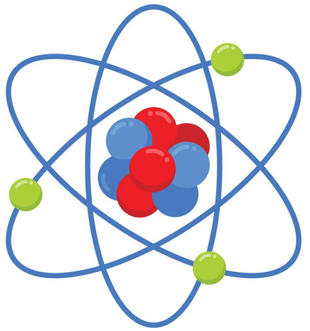
Cathode ray experiment

- The atom has an evenly distributed positive mass.
- The negatively charged electrons move distributed over the atom.
- The atoms are electrically neutral, but they can give up or take in electrons.



Development of the atomic model by Rutherford

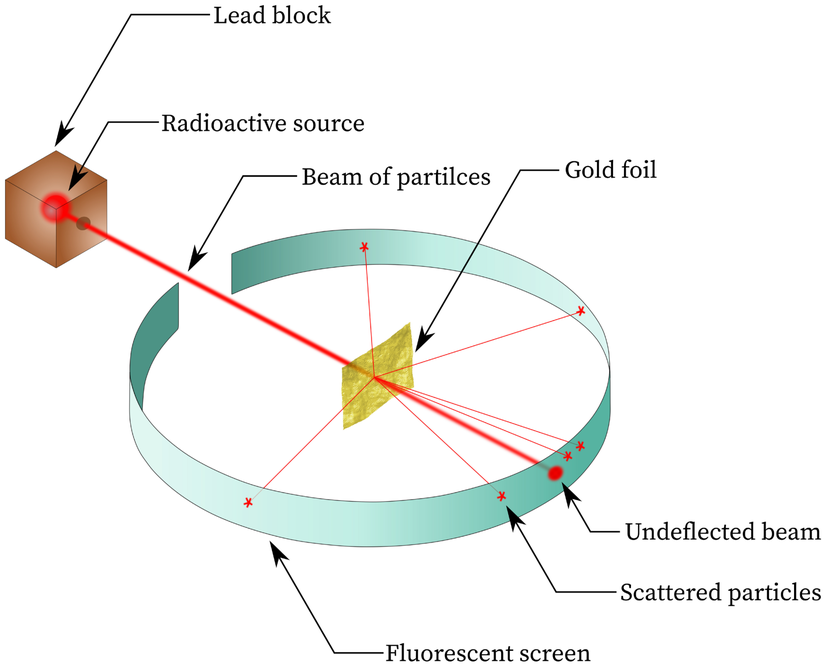
RUTHERFORD
1911

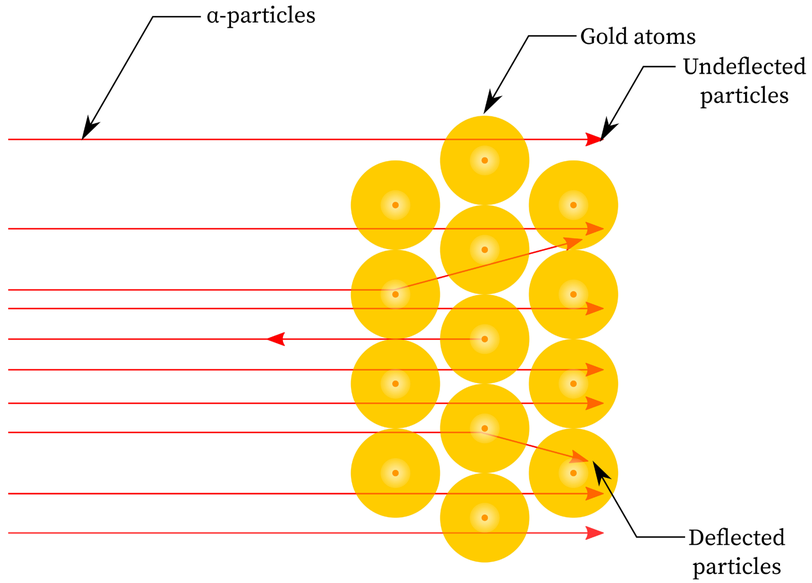


Discovered electrons surround the nucleus of an atom

Gold foil experiment

- The atom has a positive nucleus, which has a lot of mass but is very small.
- The electron shell is filled with few electrons, but determines the size of the atom.



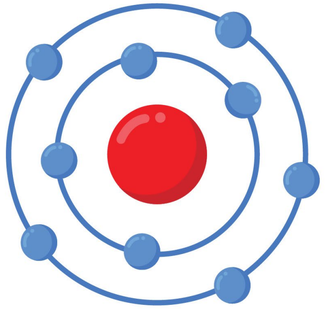


Development of the atomic model by Bohr

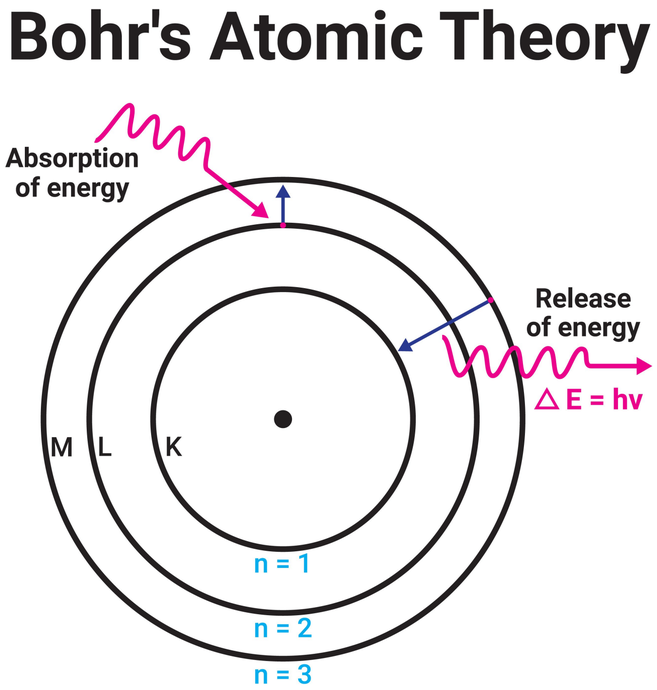
1. Bohr’s postulate: definite, stable orbits for electrons

2. Bohr’s postulate: energy of the electrons increases the farther they are from the nucleus. When they “jump” between shells, energy is absorbed or emitted (absorption or emission of a light quantum).

Shell occupancy: $2n^2$ electrons

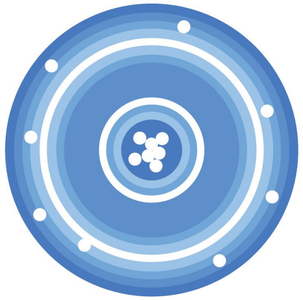


Electron move is fixed orbitals (shells)



Electron : wave or particle?

SCHRODINGER
1926



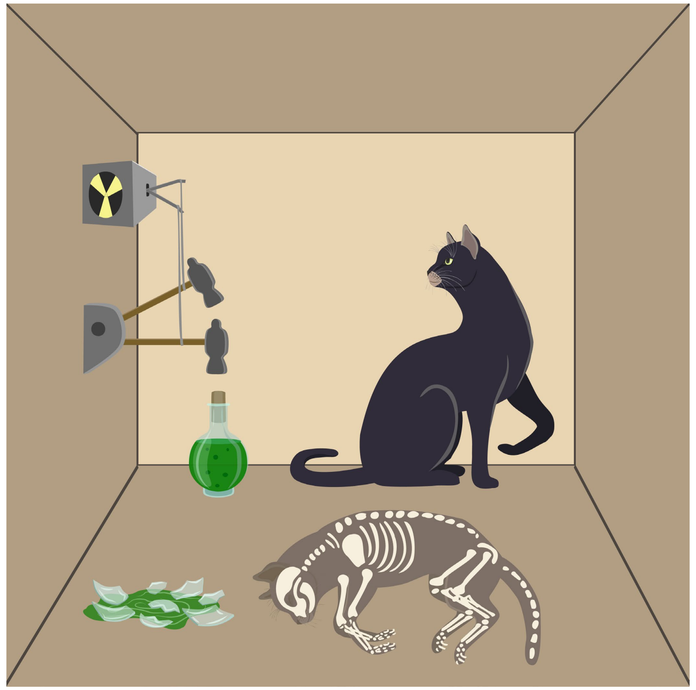
Wave behavior
of the electron

Double-slit experiment: wave nature of electrons

Heisenberg uncertainty principle: You can determine where an electron is on the shell, but not when. Or vice versa.

Schrödinger’s cat: Any manipulation of the system changes the quantum state of the cat (e.g., opening the lid), explains the probability distribution

→ Schrödinger’s equation describes the probability of finding electrons in orbitals

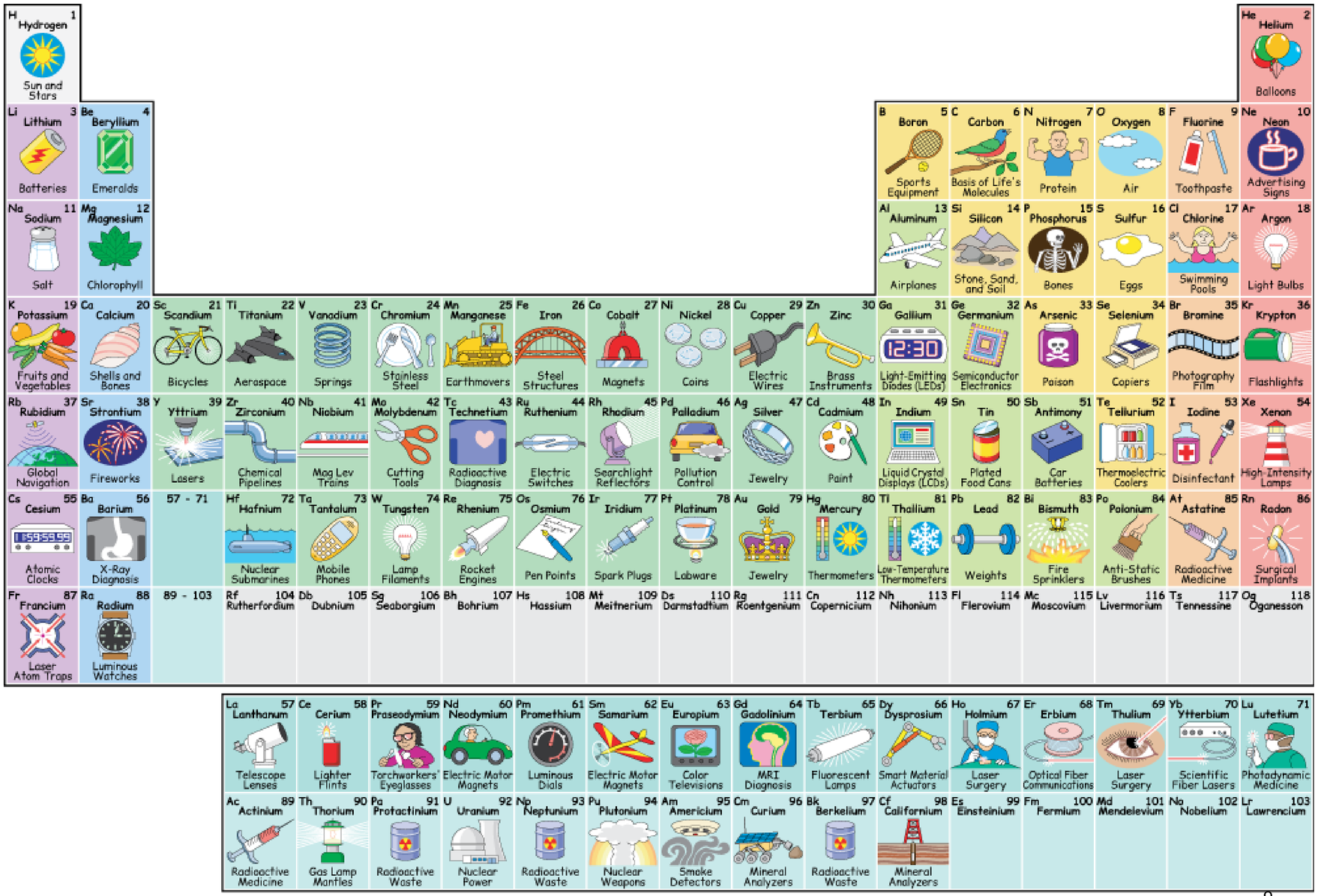


HSLU Hochschule Luzern

SW3:
The Periodic Table of the Elements

Chemistry

Dr. Macarena San Martin Ruiz
HS26



Learning objectives

Know the organization of the periodic table and describe periodic trends in metallic character, atomic radius, electron configuration, electronegativity and oxidation number.

Explain the relationship between groups and valence electrons, and chemical properties.

Locate the alkali metals, alkaline-earth metals, halogens, and noble gases in the periodic table, and describe their key properties.

Determine the electron configuration of an element using the periodic table.

Define oxidation number and identify common ones for elements using the periodic table.

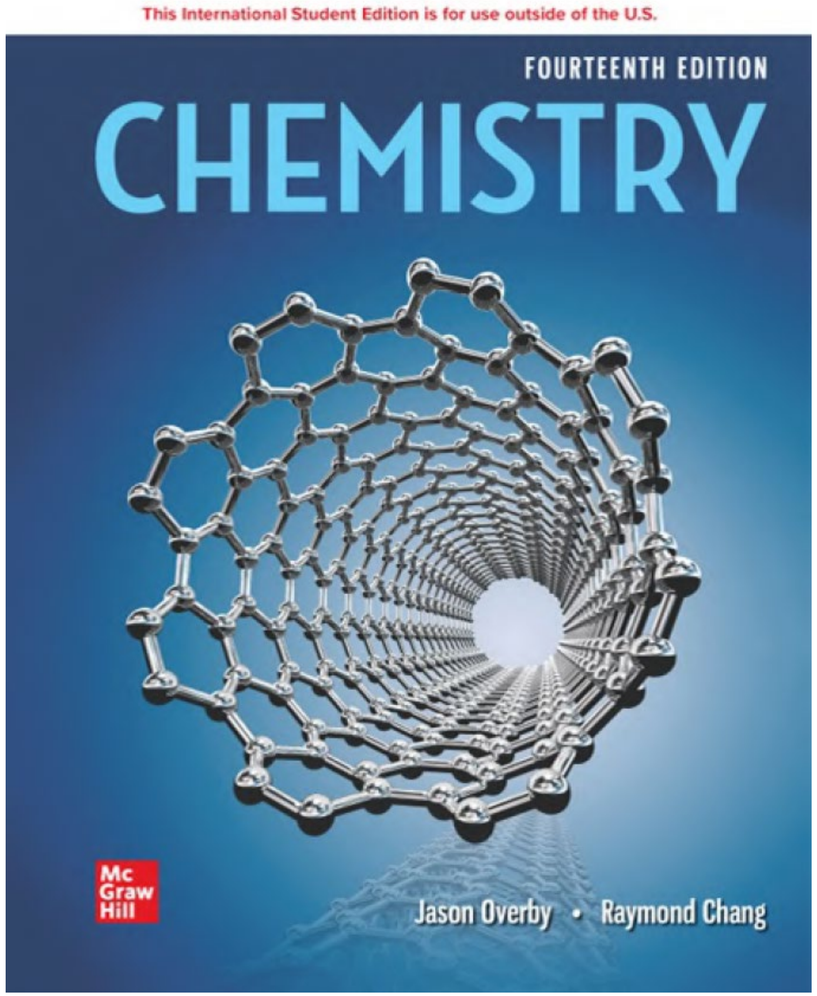
Name major elements in the cosmos, Earth’s crust and human body.

Self-study

- Video – The Periodic Table: Atomic Radius, Ionization Energy, and Electronegativity: https://www.youtube.com/watch?v=hePb00CqvP0
- Video – Trends in the Periodic Table — Reactivity: https://www.youtube.com/watch?v=Q4rW0fvkt2g
- Video – Trends in the Periodic Table — Part 2: Halogens!: https://www.youtube.com/watch?v=lizc_ehqHAs

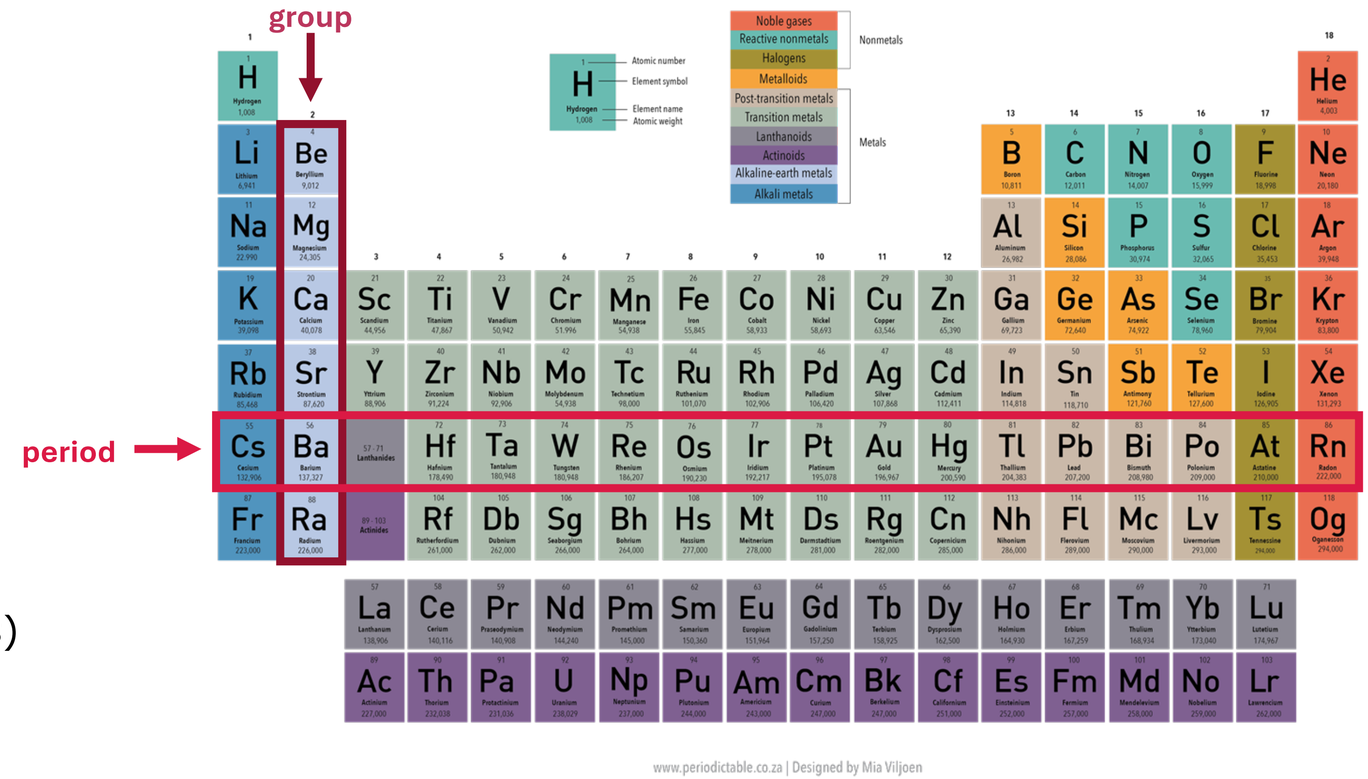
Chapters recommended

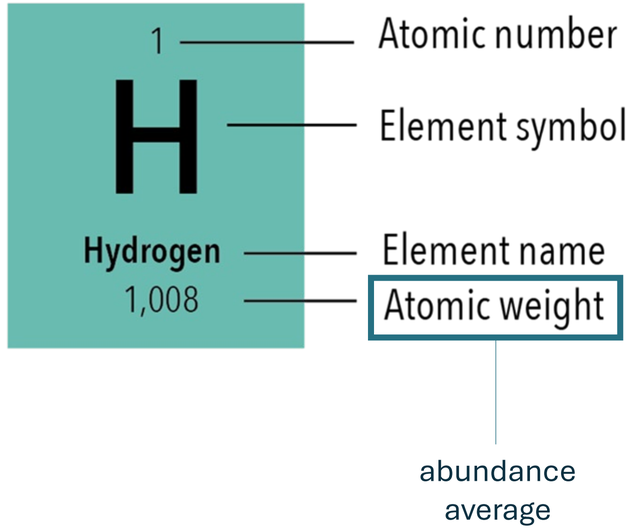
- Chapter 2: Atoms, Ions, and Molecules
  - 2.4 The Periodic Table
- Chapter 8: Periodic Relationships Among the Elements
  - 8.1 Development of the Periodic Table
  - 8.2 Periodic Classification of the Elements
  - 8.3 Periodic Variation in Physical Properties
  - 8.6 Variation in Chemical Properties of the Representative Elements
- Chapter 21: Metallurgy and the Chemistry of Metals
  - 21.4 Periodic Trends in Metallic Properties
  - 21.5 The Alkali Metals
  - 21.6 The Alkaline Earth Metals
- Chapter 22: Nonmetallic Elements and Their Compounds
  - 22.1 General Properties of Nonmetals
  - 22.6 The Halogens



Introduction

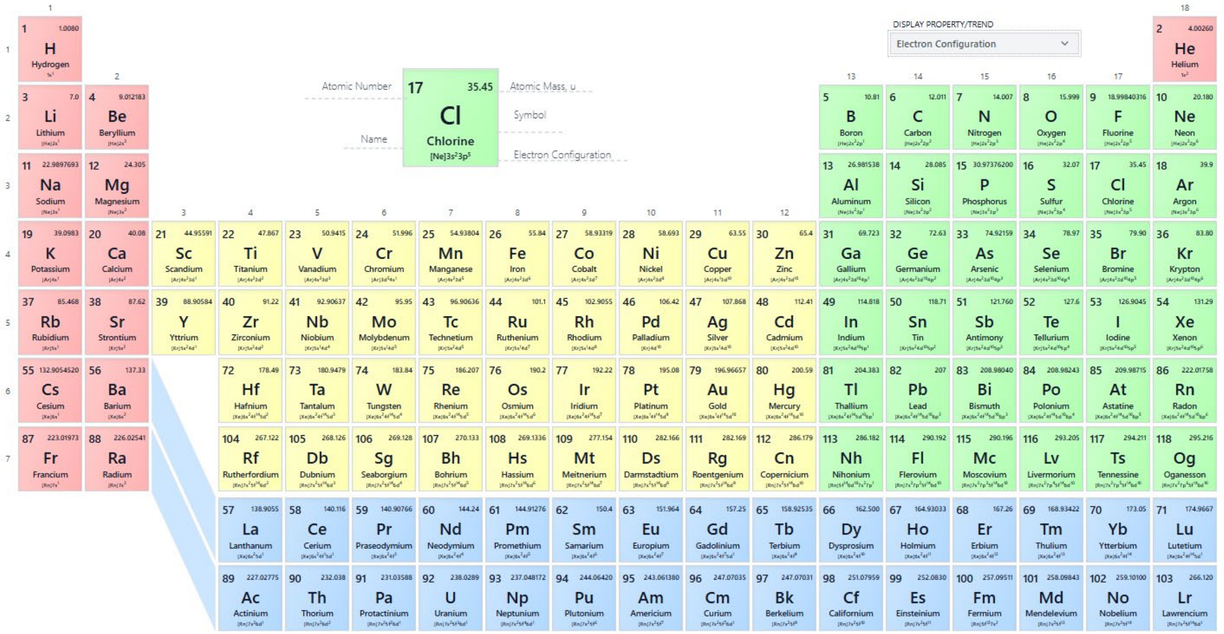
- All existing elements are displayed in the periodic table
- Systematic arrangement of elements according to chemical and physical properties
- Organised into **periods** (rows) and **groups** (columns)
- Elements arranged in order of **increasing atomic number (Z)**
- Each element is shown in a block
- Element block information:
  - Atomic number, symbol, name of element & atomic weight
  - Other possible information: oxidation numbers, radioactivity, electronegativity, electron configuration, state, density, melting point, etc.
- Important to check periodic table's legend

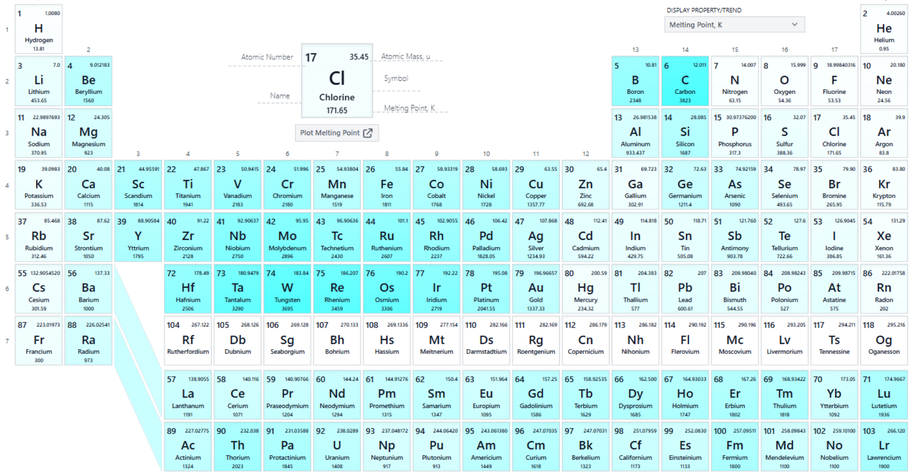


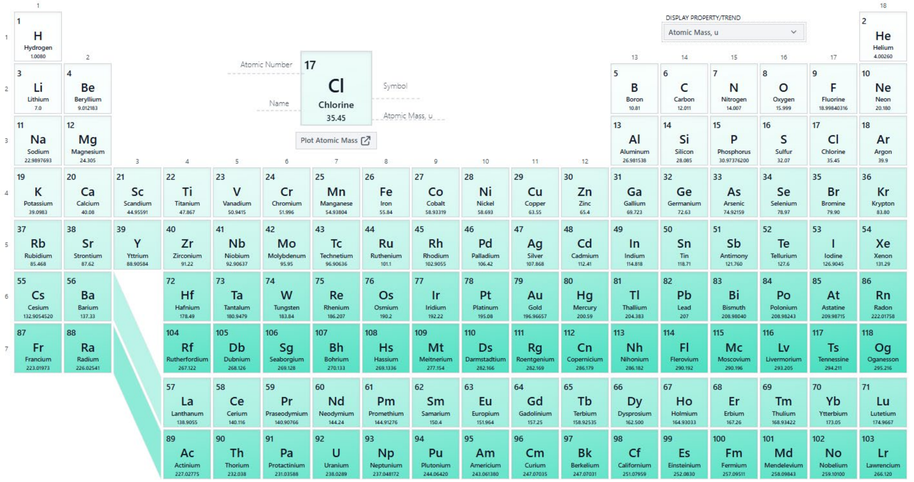


Where can I find detailed information about the periodic table and its elements?

- Periodic table by property/trend and detailed chemical element data: https://pubchem.ncbi.nlm.nih.gov/periodic-table/







Metallic character trend

- **Metals**
  - Make up >70% of elements
  - Found on the left side of the table
  - Few valence electrons, held loosely by the nucleus
  - Conduct electricity and heat well
  - Shiny, malleable and ductile
- **Nonmetals**
  - Found on the right side of the table
  - Many valence electrons, held strongly by the nucleus
  - Poor conductors of heat and electricity
- **Metalloids**
  - Located in the transition zone between metals and nonmetals
  - Properties are intermediate between the two
  - Intermediate number of valence electrons, held with medium strength

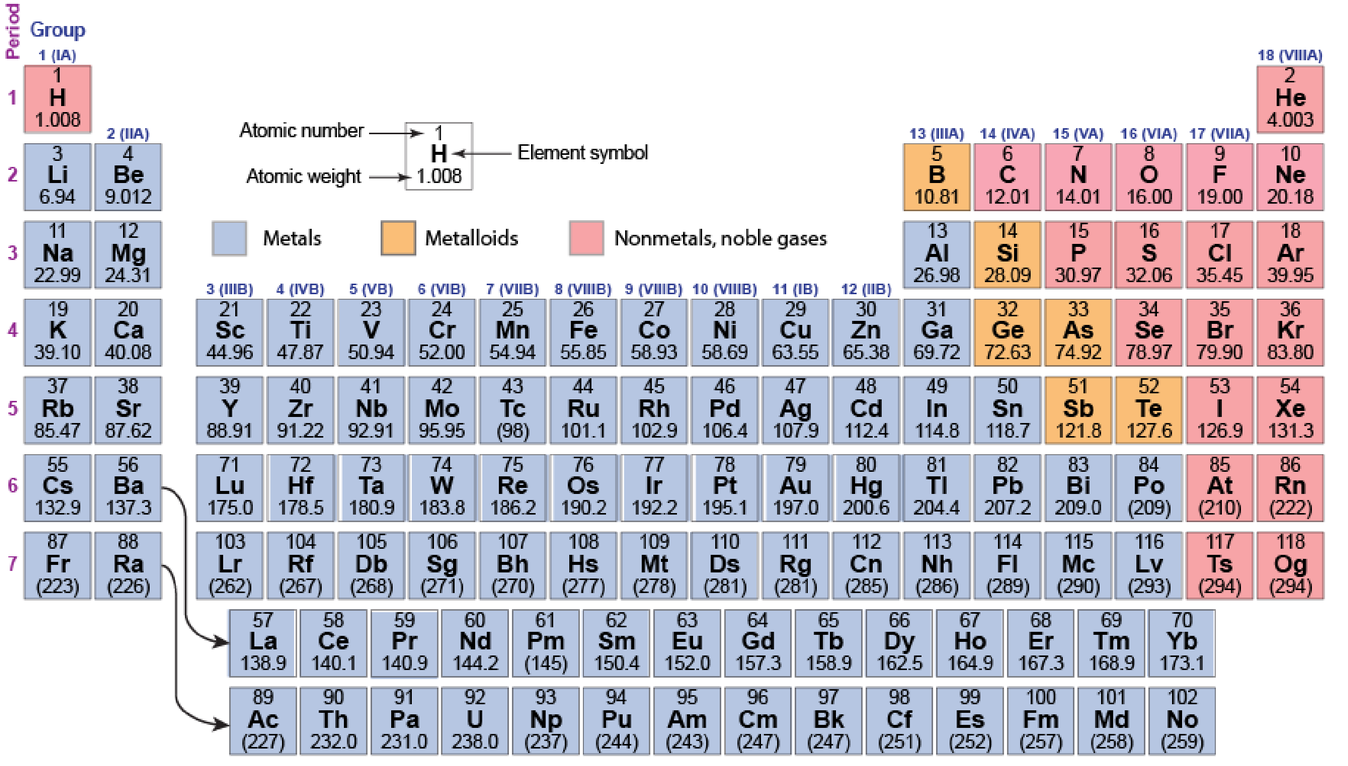
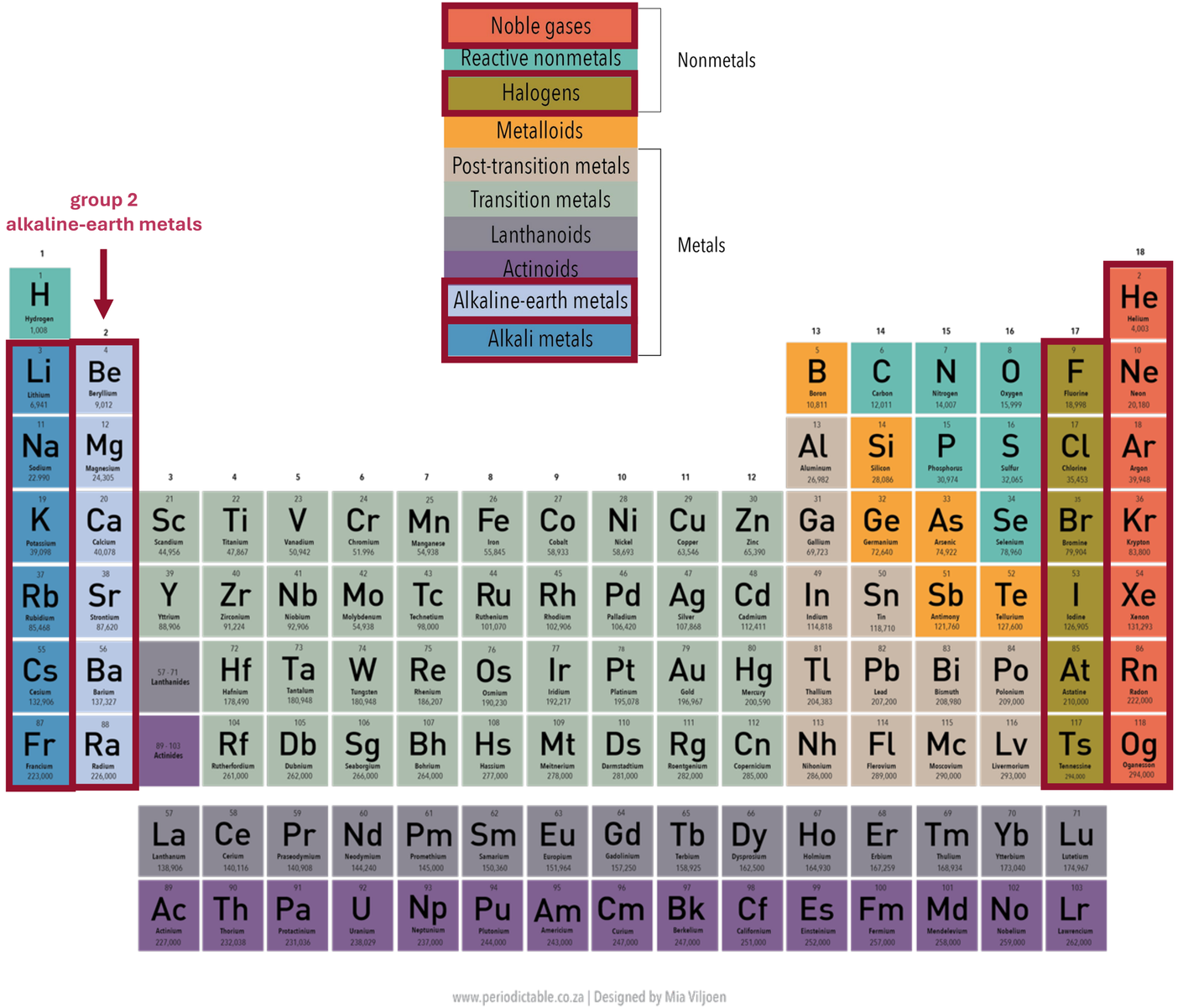


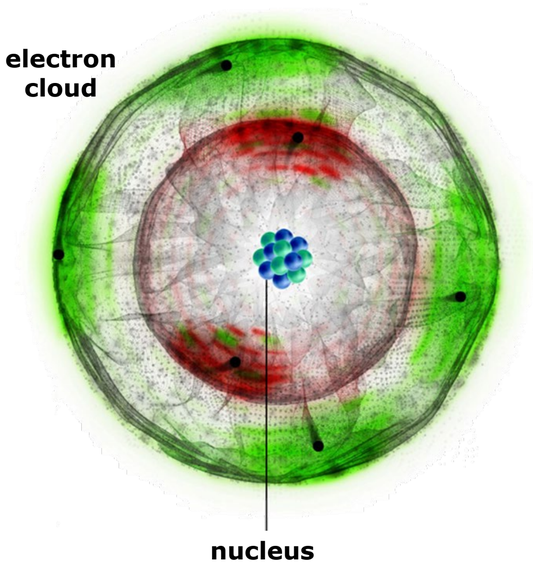
Image source: https://wisc.pb.unizin.org/chem109fall2021ver02/chapter/the-periodic-table/

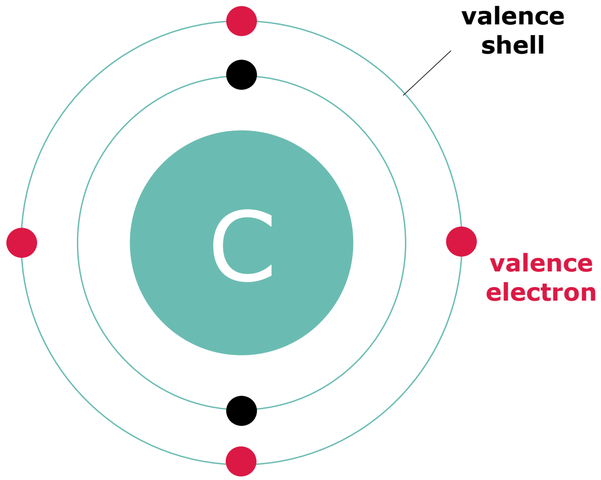
Groups in the Periodic Table

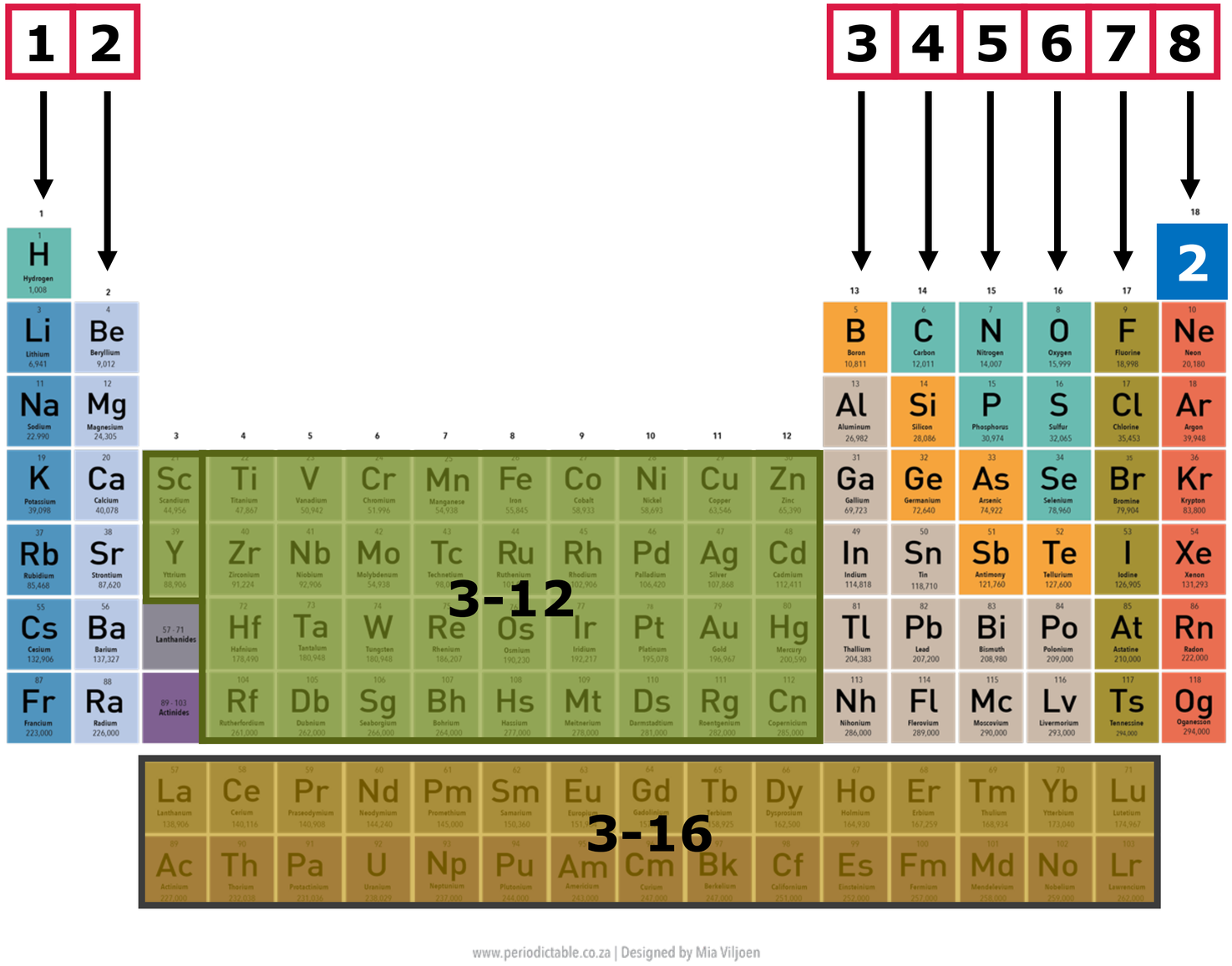
- Vertical columns
- Contain chemical-like elements with similar properties
- Help predict chemical properties of unknown elements based on well known ones in the same group
- Numbered 1 to 18 (IUPAC system); old tables use different notations (for instance 8 instead of 18)
- Main groups have specific names
  - Group 1: alkali metals
  - Group 2: alkaline-earth metals
  - Group 17: halogens
  - Group 18: noble gases



Groups and Valence Electrons



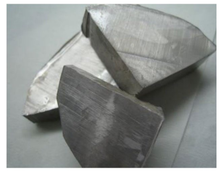




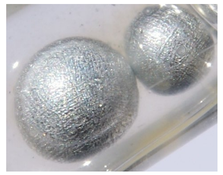
Group 1: Alkali Metals



3
Li
Lithium
6,941



11
Na
Sodium
22.990



19
K
Potassium
39,098



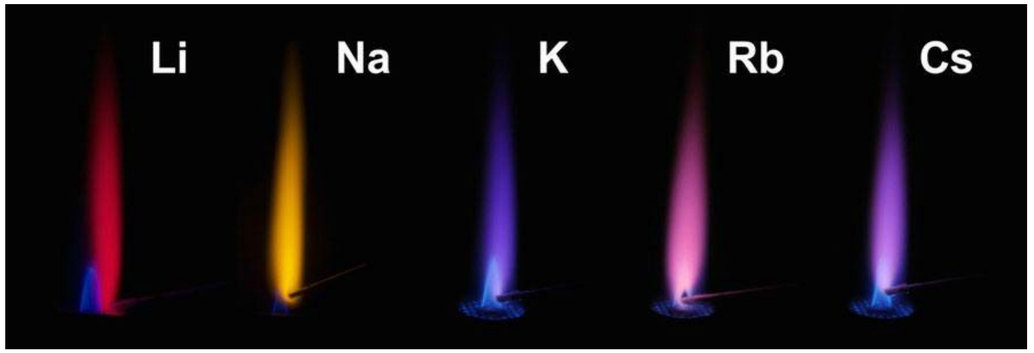
37
Rb
Rubidium
85,468



55
Cs
Cesium
132,906

87
Fr
Francium
223,000

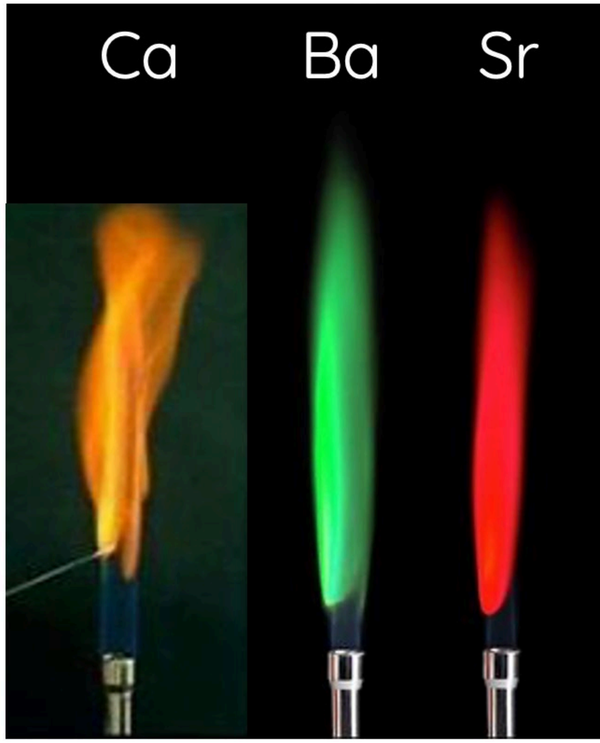
- Shiny, silvery and soft (can often be cut with a knife)
- 1 valence electron → highly reactive
- Reactivity increases down the group (atomic size gets larger, electron is lost more easily)
- Very low electronegativity
- Strong reaction with water, explosive for heavier elements (Cs, Fr): https://www.youtube.com/watch?v=jI__JY7pqOM
- Strong reducing agents (easily give up electrons)
- Special flame colour



Group 2: Alkaline-earth metals



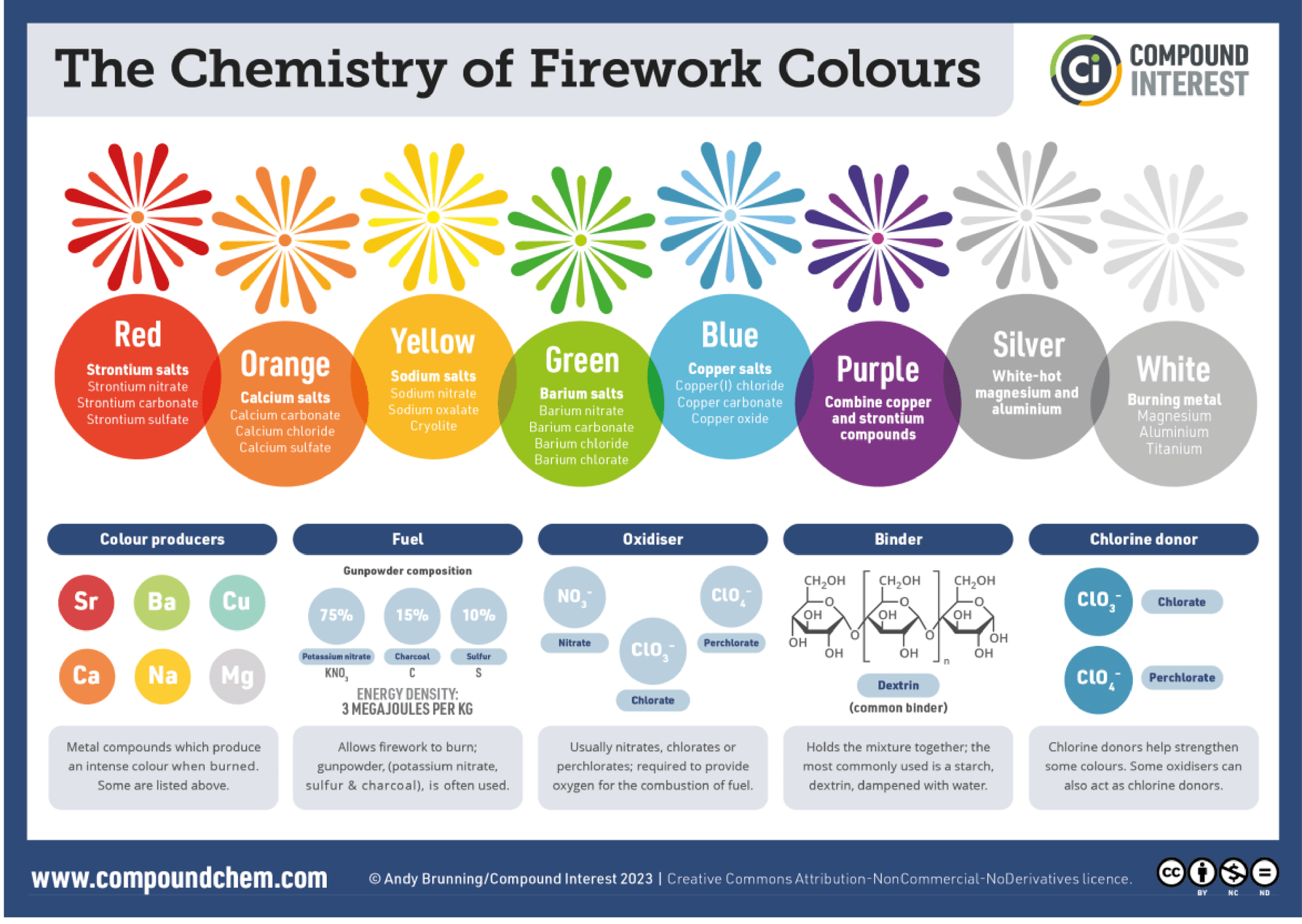
- Shiny, silvery-white
- 2 valence electrons → somewhat reactive
- Low electronegativity
- React with water, but less strongly than alkali metals
- Good reducing agents (easily give up electrons)
- Intense flame colour of Ca, Ba and Sr



Excursus

Chemistry of Fireworks

- Fireworks simulation: https://javalab.org/en/fireworks_en/



Group 17: Halogens

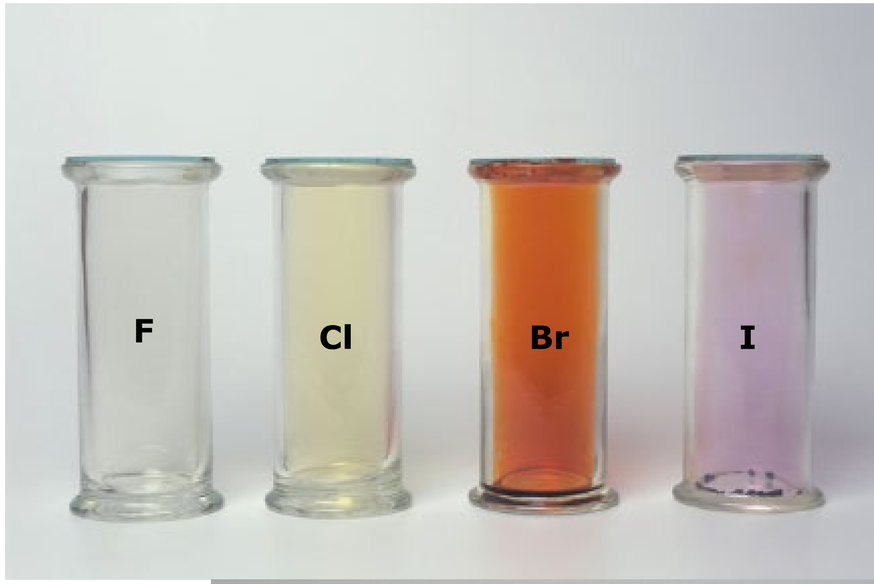
most reactive element

used as desinfectants (sterilization)

unknown chemical properties

9	17	35	53	85	117
F	Cl	Br	I	At	Ts
Fluorine	Chlorine	Bromine	Iodine	Astatine	Tennessine
18,998	35,453	79,904	126,905	210,000	294,000

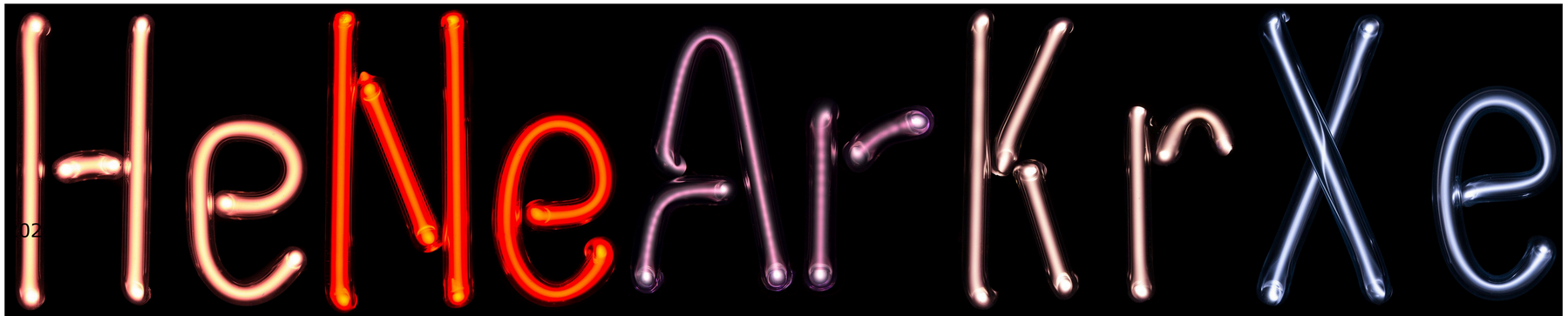
- Nonmetals with characteristic color
- Exist as diatomic molecules (F_2, Cl_2, Br_2, I_2)
- Contains elements in 3 states of matter at standard conditions (0°C, 1 atm):
  - F_2 and Cl_2 are toxic gases
  - Br_2 is liquid; I_2 is solid
- 7 valence electrons, high electronegativity → highly reactive → harmful to biological organisms
- Less reactive as atomic number increases
- Halogen means “salt maker”; react with metals to form a metal salt (for instance NaCl = common salt)
- Good oxidating agents (easily accept electrons)





Group 18: Noble gases

- Historically known as inert gases
- Odorless, colorless, monatomic gases
- Found mainly in the Earth's atmosphere (air)
- Full valence shell → very low reactivity → useful when you want to avoid unwanten chemical reactions
- Make lights under electricity application (neon lamp)
- Application
  - Welding: used as protective gas to protect against oxidation

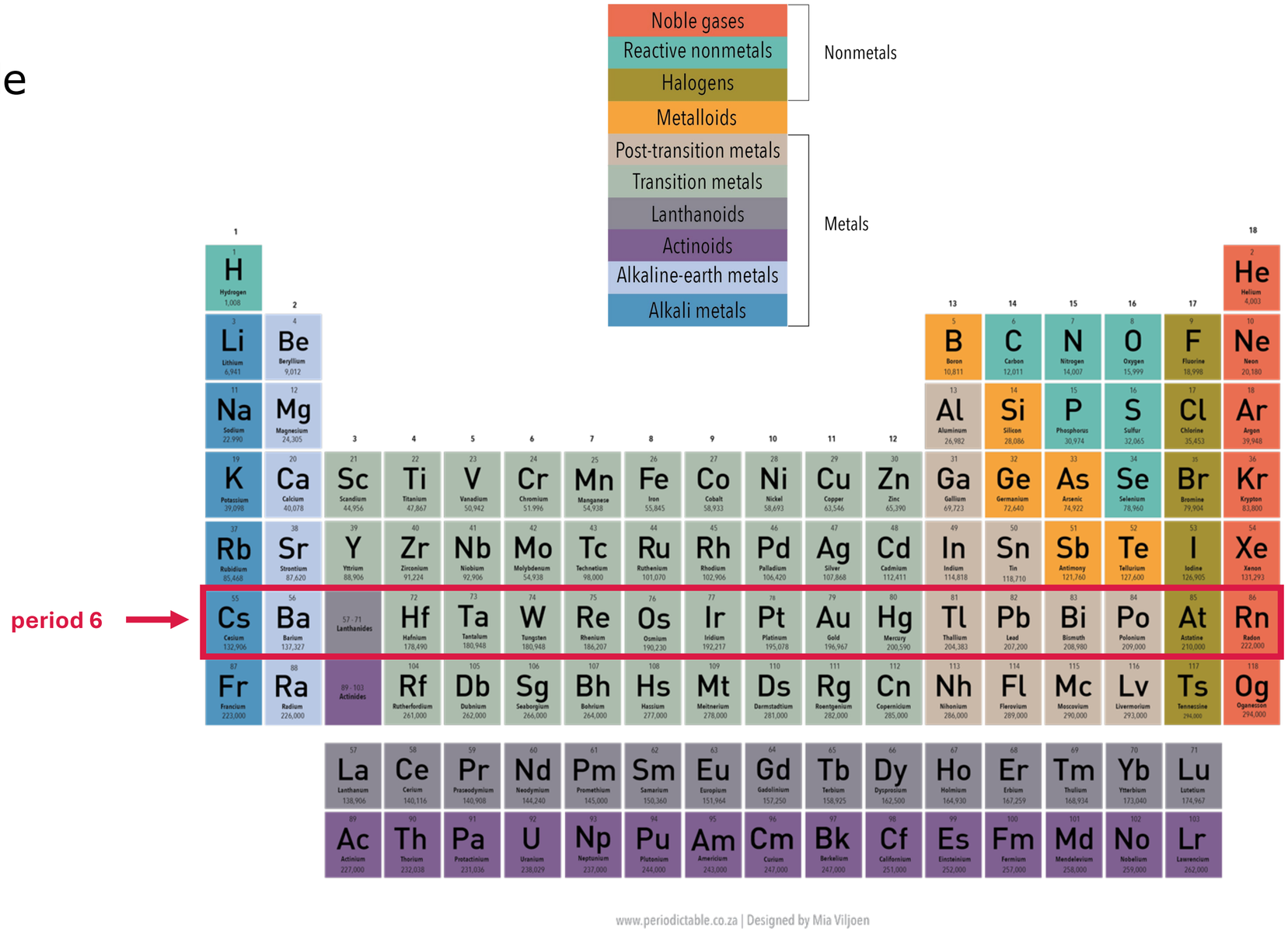


Excursus: Extra videos

- Reaction of Alkali Metals with water: https://www.youtube.com/watch?v=eaChisV5uR0
- The rainbow flame demonstration: https://www.youtube.com/watch?v=TOyDzOc2AaI
- Reaction of halogens with sodium: https://www.youtube.com/watch?v=saLvwX3_p1s

Periods in the Periodic Table

- Horizontal rows
- Numbered from 1 to 7



Group exercise

Which element drawn in our periodic table has the biggest atomic radius?

Why?

Group exercise

Which element drawn in our periodic table has the biggest atomic radius?

Why?

Elements shown on our periodic table on the blackboard: Fe, Li, Si, O, Ge, Be, F, K, Ne

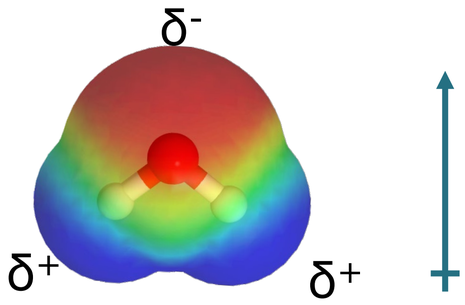
Atomic radius (largest → smallest): K > Ge > Fe > Si > Li > Be > O > F > Ne

Potassium (K) has the largest atomic radius because:

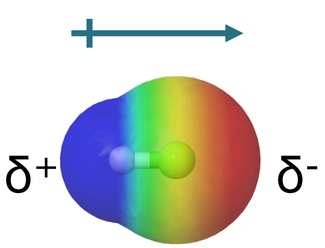
- Atomic radius increases down a group (more electron shells)
- Atomic radius decreases across a period (greater nuclear charge pulls electrons closer)
- K is in period 4, group 1, so it has more electron shells than the others and only one valence electron, which is held loosely by the nucleus

Periodic trend in electronegativity

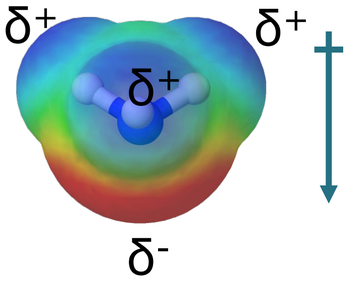
- Tendency of an atom to attract shared electrons in a bond
- Defined for atoms in molecules (not isolated atoms)
- Pauling scale is the most common measure
- The more electronegative → the more attracts electrons
- Effects of electronegativity: polarity of chemical bonds, polarity of molecules



water (H_2O)



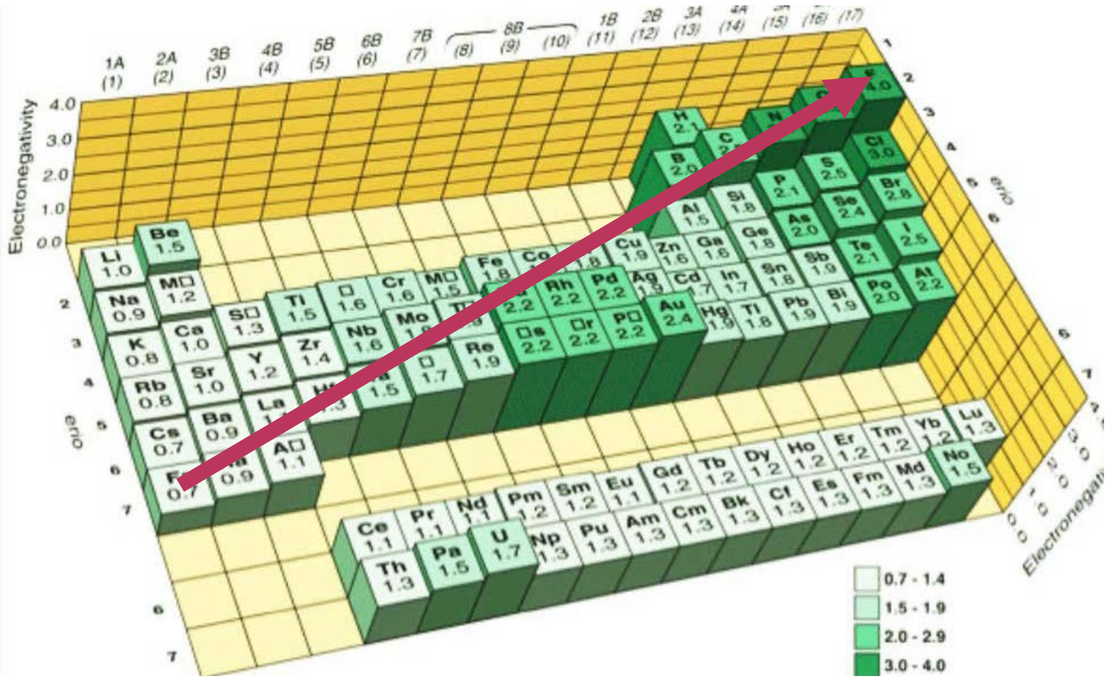
hydrogen fluoride (HF)

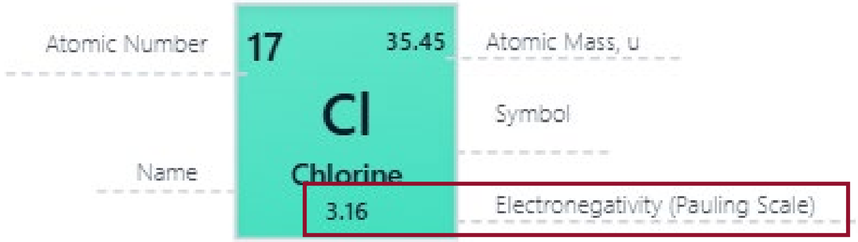


ammonia (NH_3)

Periodic trend in electronegativity

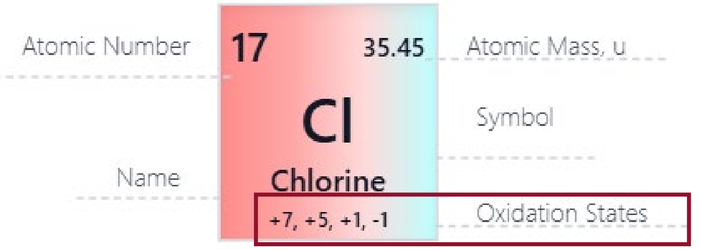
- From left to right across the period table electronegativity increases
  - Increases with nuclear charge (more protons = stronger pull)
- From top to bottom electronegativity decreases
  - Decreases when valence electrons are farther from the nucleus (more shells, more shielding → larger atomic size)

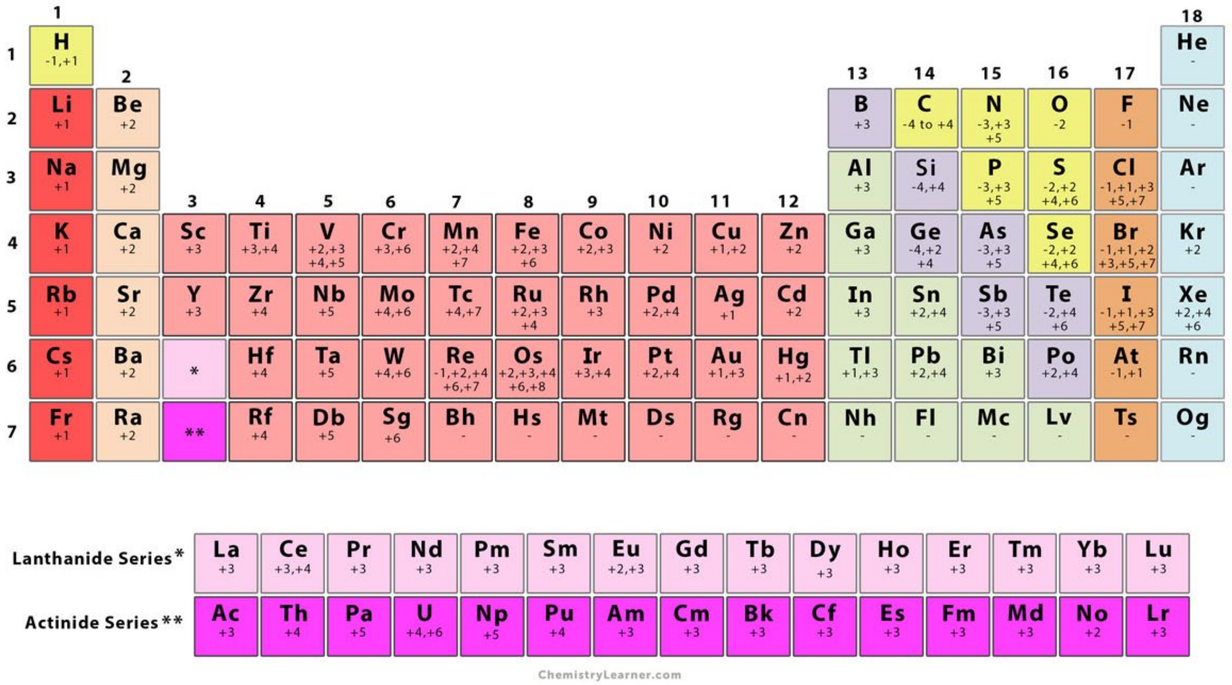




Periodic trend in oxidation numbers

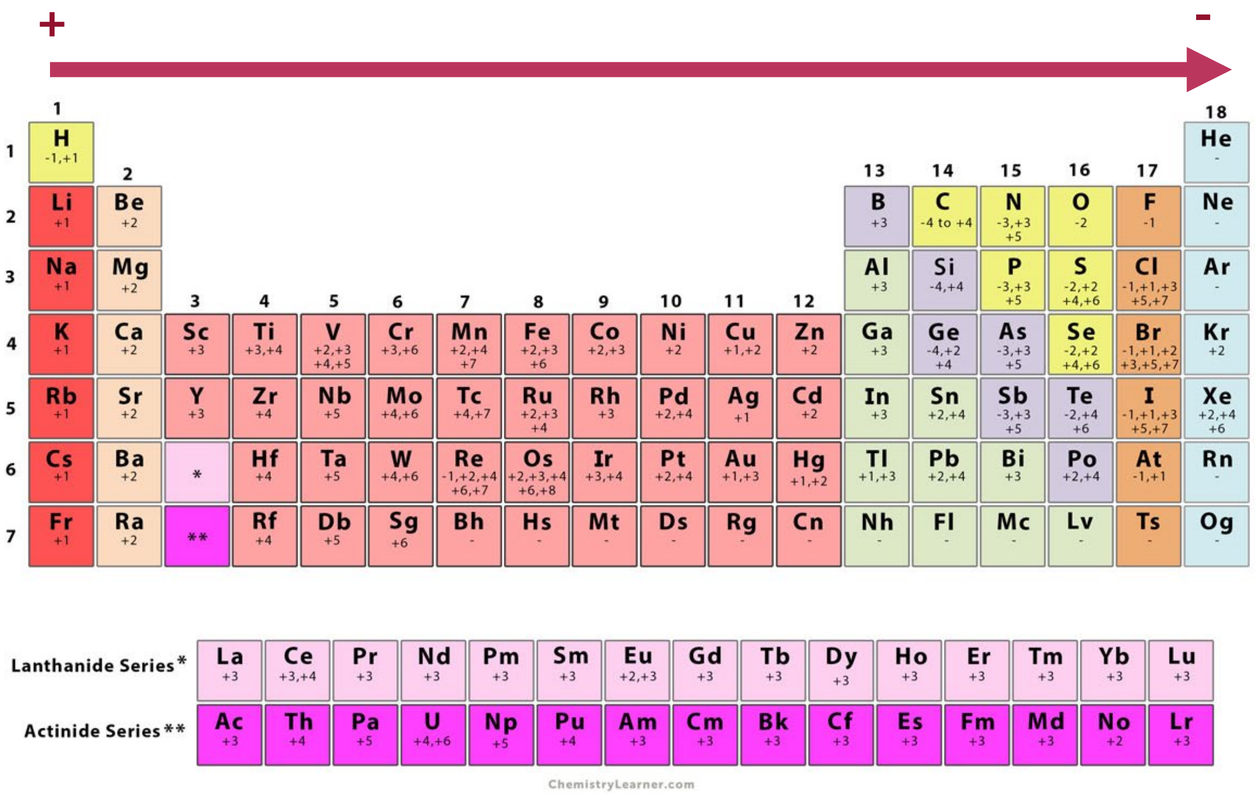
- Oxidation number/oxidation state (OS): hypothetical charge of an atom if all of its bonds to other atoms would be fully ionic
- Describes the degree of oxidation (loss of electrons) of an atom in a chemical compound
  - Positive (release of electrons), negative (electrons gain) or zero
  - Usually written in roman numerals (+I, +II, -III, -IV, etc.)
- Useful when working with redox (reduction-oxidation) reactions





Periodic trend in oxidation numbers

- The sum of OS in a neutral molecule (no charge) is always 0; in an ion, equals the total charge of the ion
- **Key oxidation states**:
  - Element in free form: 0
  - H: +1 or -1
  - O: -2
  - Rest of s-block: group nr. = OS
    - Alkali (group 1): +1
    - Alkaline-earth metals (group 2): +2
  - Halogens (group 17): commonly -1
- P-block elements: multiple, variable OS (ns and np valence electrons → more opportunities for bonding)
  - Group 13 ($ns^2\ np^1$): +3 (and +1 in case of Tl)
  - Group 14 ($ns^2\ np^2$): +2, +4
  - Group 15 ($ns^2\ np^3$): -3, +3, +5
  - Group 16 ($ns^2\ np^4$): -2, +2, +4, +6
  - Group 17 ($ns^2\ np^5$): -1
- D-block: wide range of stable oxidation states



Exercise

Identify the most common oxidation states of these elements; write them in the periodic table (blue=metal; red=nonmetal; white=metalloid)

14. Be:

15. B:

16. C:

17. N:

18. O:

19. F:

Exercise

Identify the most common oxidation states of these elements; write them in the periodic table (blue=metal; red=nonmetal; white=metalloid)

14. Be: Beryllium, Z=4, group 2, period 2, metal, OS: +2

15. B: Boron, Z=5, group 13, period 2, metalloid, OS: +3

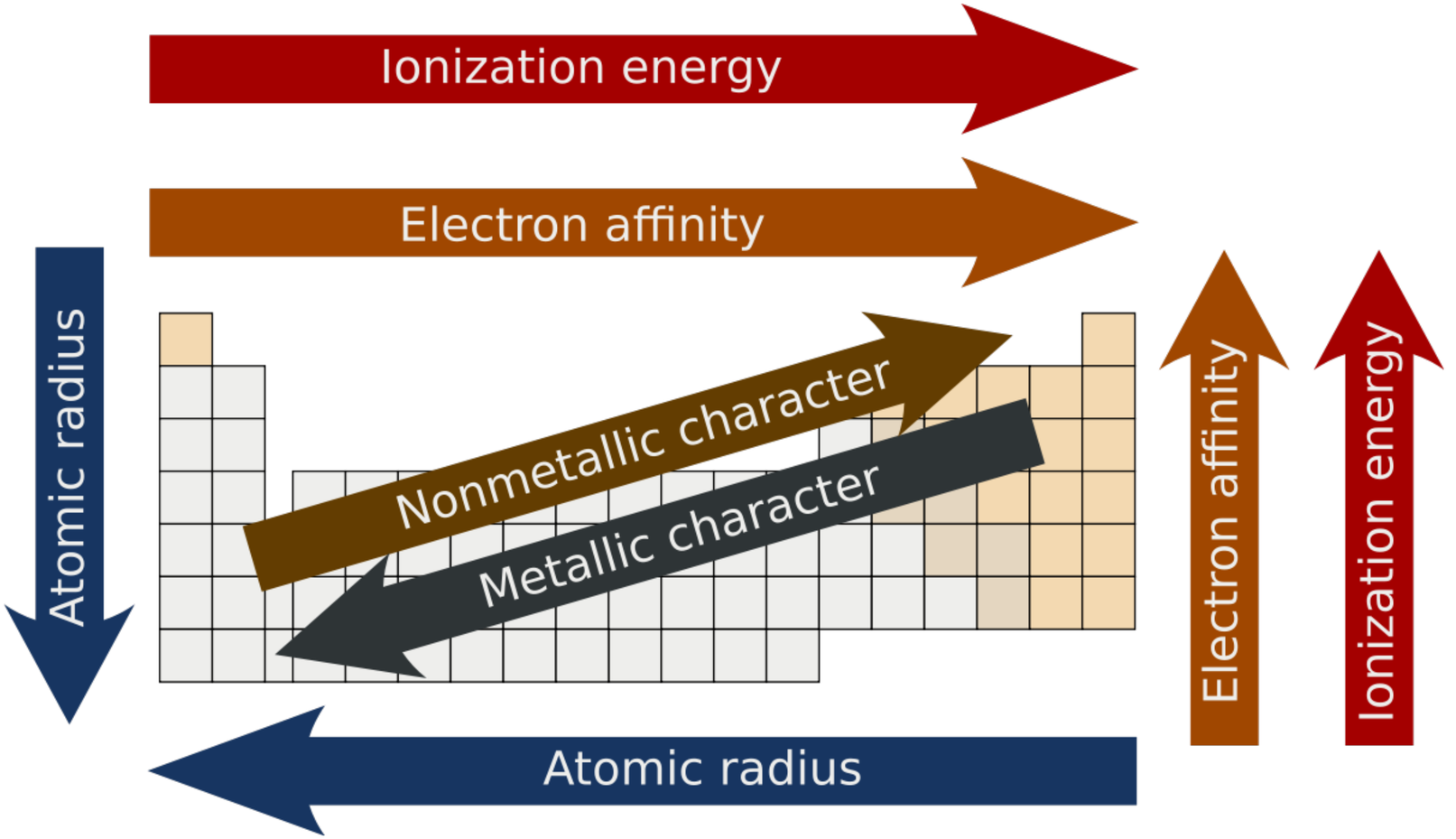
16. C: Carbon, Z=6, group 14, period 2, nonmetal, OS: -4, +2, **+4 (most common)**

17. N: Nitrogen, Z=7, group 15, period 2, nonmetal, OS: -3, +3, +5

18. O: Oxygen, Z=8, group 16, period 2, nonmetal, OS: -2

19. F: Fluorine, Z=9, group 17, period 2, nonmetal, OS: -1

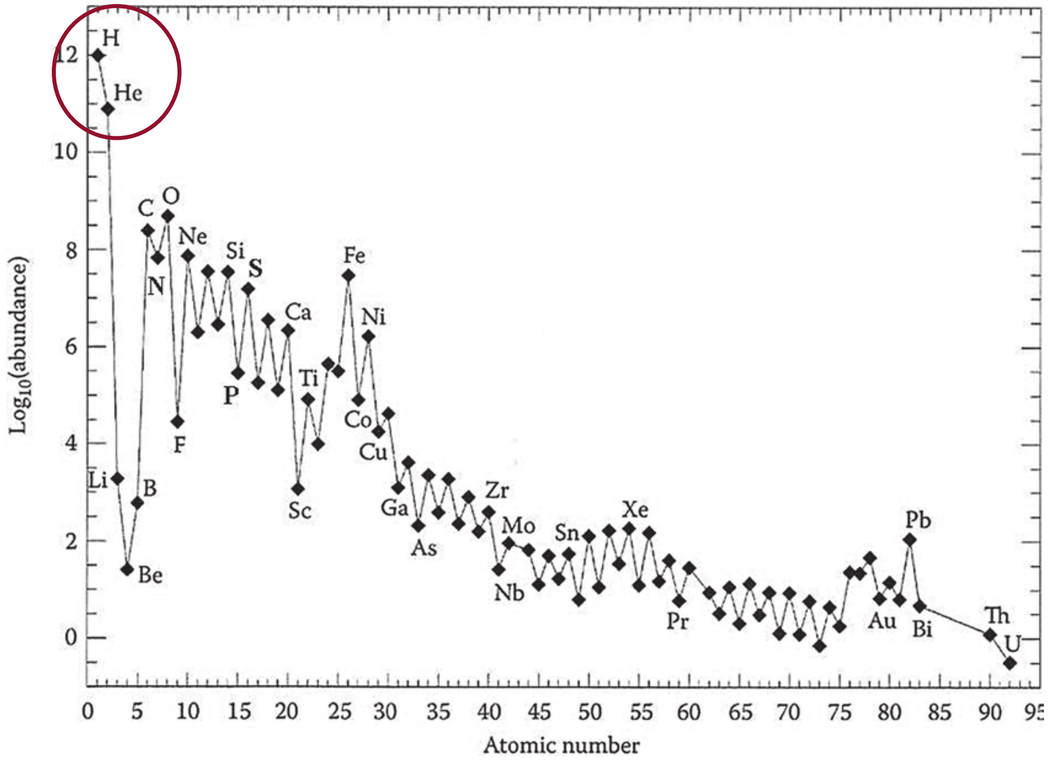
Summary of periodic trends



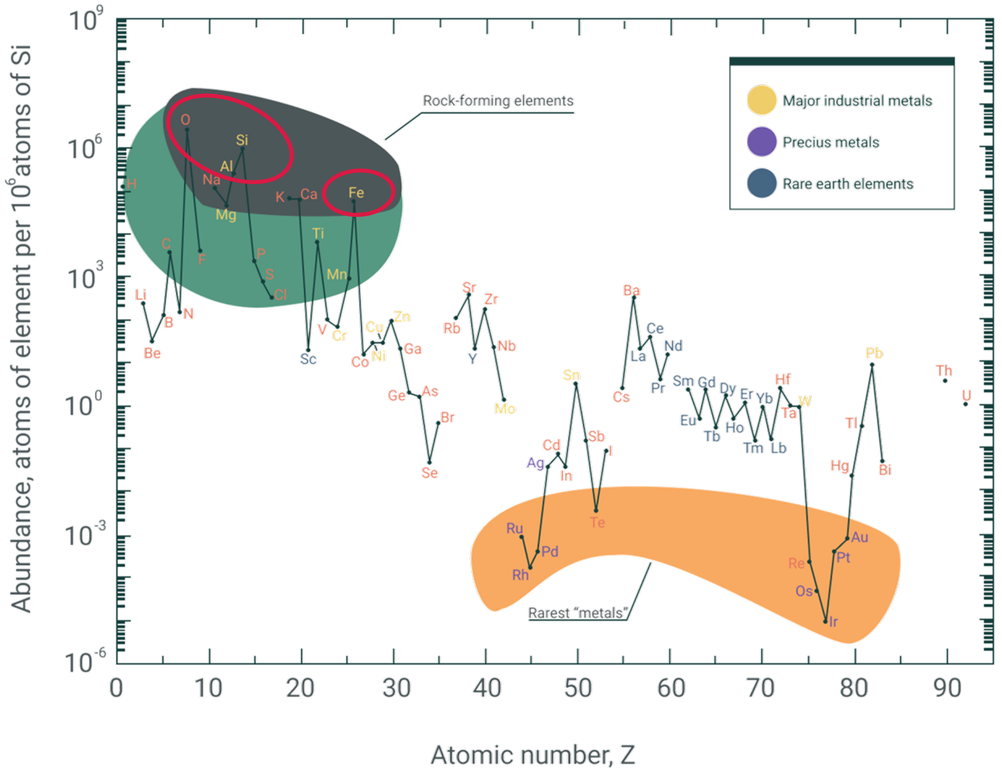
39

Abundance of elements

Cosmos



Earth’s crust (surface to a depth of about 40 km)



Abundance of elements

Human body

Element	Percent by Mass*	Biological Relevance
Oxygen (O)	65%	Component of water and organic molecules
Carbon (C)	18%	Backbone of biomolecules (carbohydrates, lipids, proteins, nucleic acids)
Hydrogen (H)	10%	Part of water and organic molecules
Nitrogen (N)	3%	Proteins and nucleic acids
Calcium (Ca)	1.6%	Bone structure, signaling (Ca^{2+})
Phosphorus (P)	1.2%	DNA, ATP, phospholipids
Potassium (K)	0.2%	Electrolyte, nerve function
Sulfur (S)	0.2%	In amino acids (cysteine, methionine)
Chlorine (Cl)	0.2%	Stomach acid (HCl), water balance
Sodium (Na)	0.1%	Electrolyte, water balance, nerve conduction
Magnesium (Mg)	0.05%	Cofactor for enzymes, muscle and nerve function
Iron (Fe)	<0.05%	Hemoglobin (oxygen transport)
Cobalt (Co)	<0.05%	Part of Vitamin B12
Copper (Cu)	<0.05%	Enzyme cofactor
Zinc (Zn)	<0.05%	Enzyme cofactor, immune system
Iodine (I)	<0.05%	Thyroid hormones
Selenium (Se)	<0.01%	Antioxidant defense (glutathione peroxidase)
Fluorine (F)	<0.01%	Tooth enamel, bone strength

*Percent by mass *gives the mass of the element in grams present in a 100-g sample*

Major elements

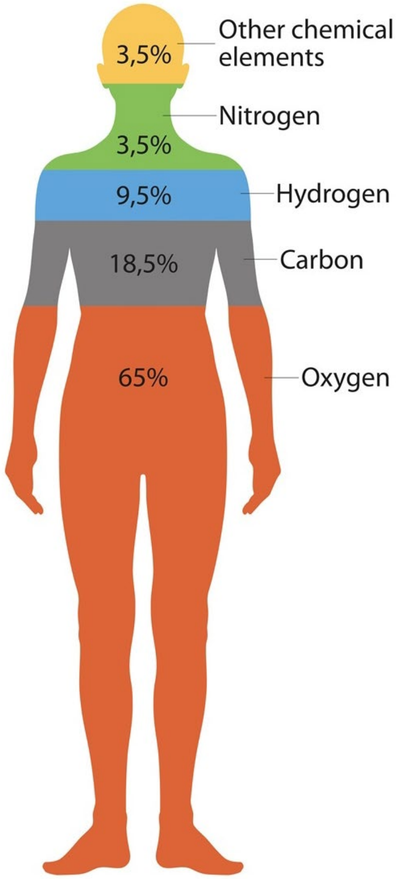
- ~96% of body mass
- Nonmetals → stable covalent bonds, low atomic weight

Macroelements
(≥ 50 mg/kg body weight)

- Usually present as ions in aqueous medium (Na^+, K^+, Ca^{2+}, Cl^-, SO_4^{2-}, etc.)
- Essential as electrolytes → nerve signaling, osmosis

Trace elements
(< 50 mg/kg body weight)

- Essential in small amounts
- Imbalances (deficiency or excess) can cause serious illness, developmental problems, or even death
- Common deficiencies: iron (anemia), iodine (thyroid disorders)



Questions?



Exercises: Tasks 1–9

More exercises: self-study

1. Explain why atomic radius decreases across a period.
2. Which element is more electronegative: N or O? Explain your reasoning.
3. Explain how oxidation states change from left to right across Period 3
4. What are the most common oxidation states for alkali, alkaline-earth metals and halogens? Why?
5. Fill in the oxidation numbers for the elements below in their common compounds:
   - Fe in Fe_2O_3
   - Mn in $KMnO_4$
   - S in H_2SO_4
   - Cl in $HClO_3$

More exercises: self-study - Answers

1. Explain why atomic radius decreases across a period. *Atomic radius decreases across a period because the number of protons increases, creating a stronger nuclear charge that pulls the electrons closer to the nucleus.Even though more electrons are added, they enter the same energy level, so shielding doesn't increase much, and the atoms become smaller from left to right.*

2. Which element is more electronegative: N or O? Explain your reasoning. *Oxygen is more electronegative. Both are in Period 2, and electronegativity increases left → right; oxygen has a higher effective nuclear charge and a smaller atomic radius than nitrogen, so it pulls shared electrons more strongly (Pauling: O ≈ 3.44, N ≈ 3.04).*

3. Explain how oxidation states change from left to right across Period 3. *Across Period 3, oxidation states change from positive to negative — metals on the left (like Na, Mg, Al) lose electrons and have positive oxidation states, while nonmetals on the right (like S, Cl) gain electrons and have negative oxidation states.*

4. What are the most common oxidation states for alkali, alkaline-earth metals and halogens? Why? *Alkali metals: +1; alkaline-earth metals: +2; halogens: –1; Alkali and alkaline-earth metals have 1 or 2 valence electrons, which they easily lose to reach a stable noble gas configuration. Halogens have 7 valence electrons, so they gain 1 electron to complete their outer shell and become stable.*

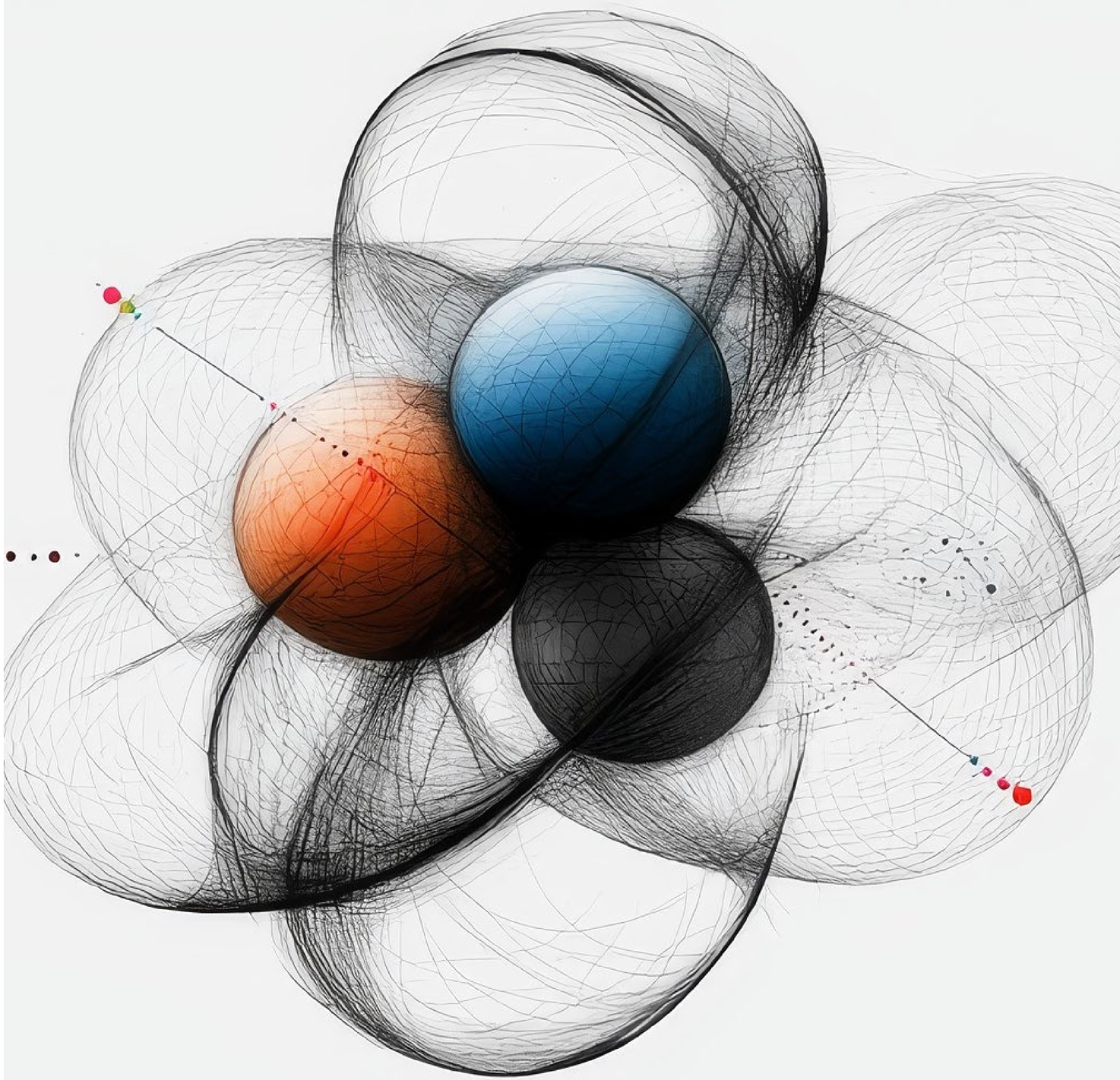
5. Fill in the oxidation numbers for the elements below in their common compounds:
   - Fe in Fe_2O_3: *O: -2; total=-6;* ***Fe: +3***
   - Mn in $KMnO_4$: *K:+1; O:-2;* ***Mn:+7***
   - S in H_2SO_4: *H: +1; O: -2;* ***S: +6***
   - Cl in $HClO_3$: *H:+1; O: -2;* ***Cl: +5***

HSLU Lucerne University of Applied Sciences and Arts

Electronic configuration

Dr. Macarena San Martin Ruiz

HS26

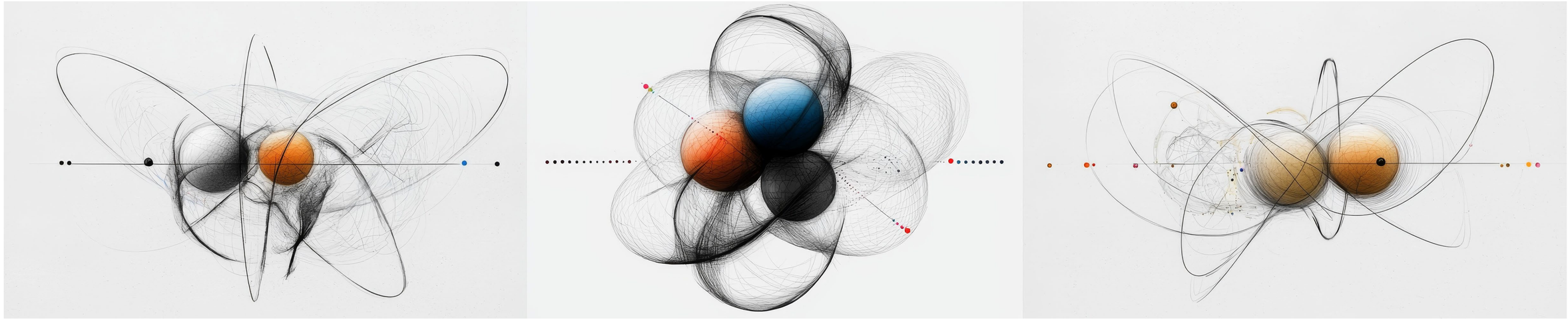


Engineering and Architecture

September 29, 2026

FH Zentralschweiz

Content



- Different atomic models
- Orbital theory

Learning objectives

Know the 4 quantum numbers and understand what they describe

Be familiar with the Pauli exclusion principle

Be able to correctly fill electrons of an element into orbitals (know electron configuration, apply Hund's rule)

Be able to determine the valence electrons of an element using the periodic table

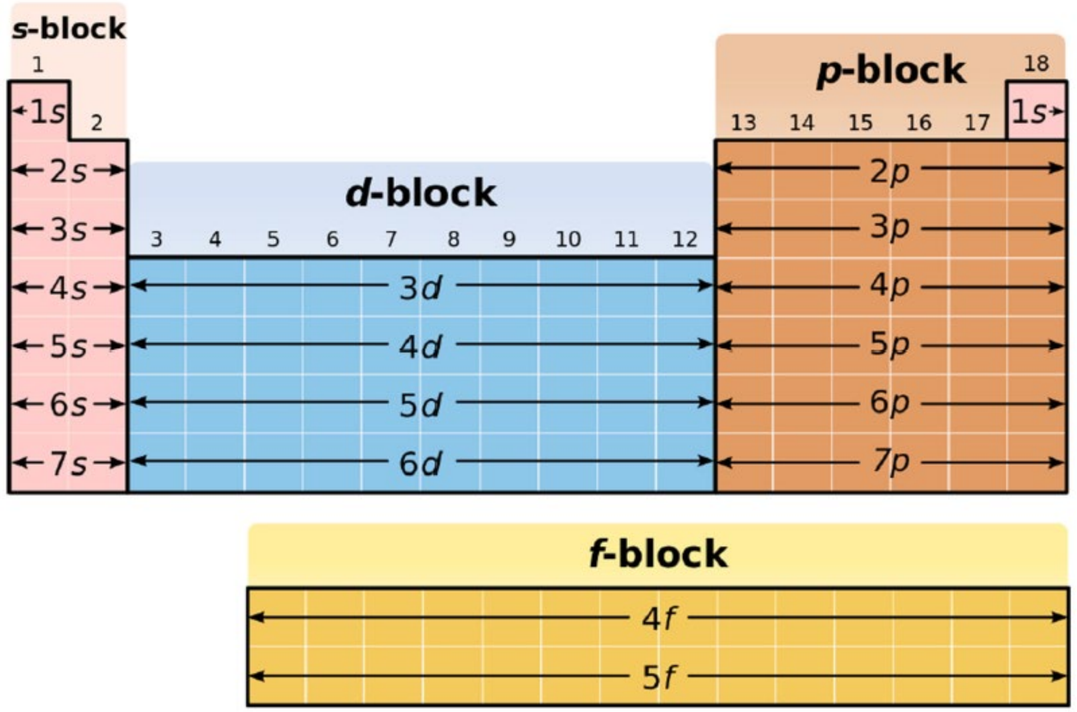
Chapters recommended

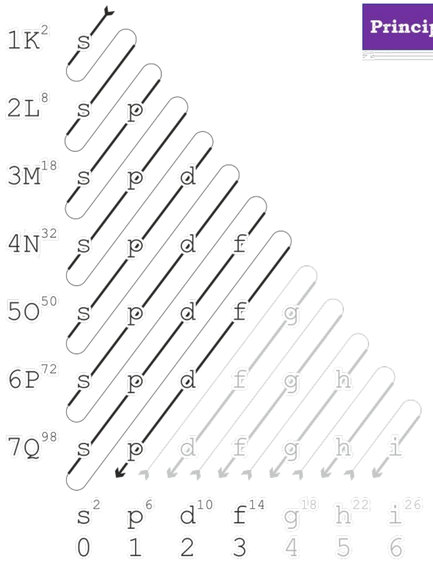
Chapter 7: closer look: 7,1,7.4, 7.3, 7.5,7.6,7.7, 7.8, 7.9, Chemistry in action (quantum dots)

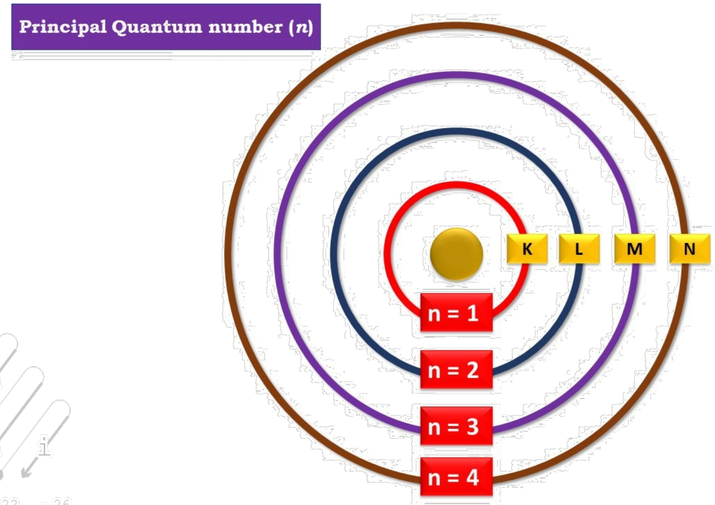
Chapter 10: all

Periodic trend in electron configuration

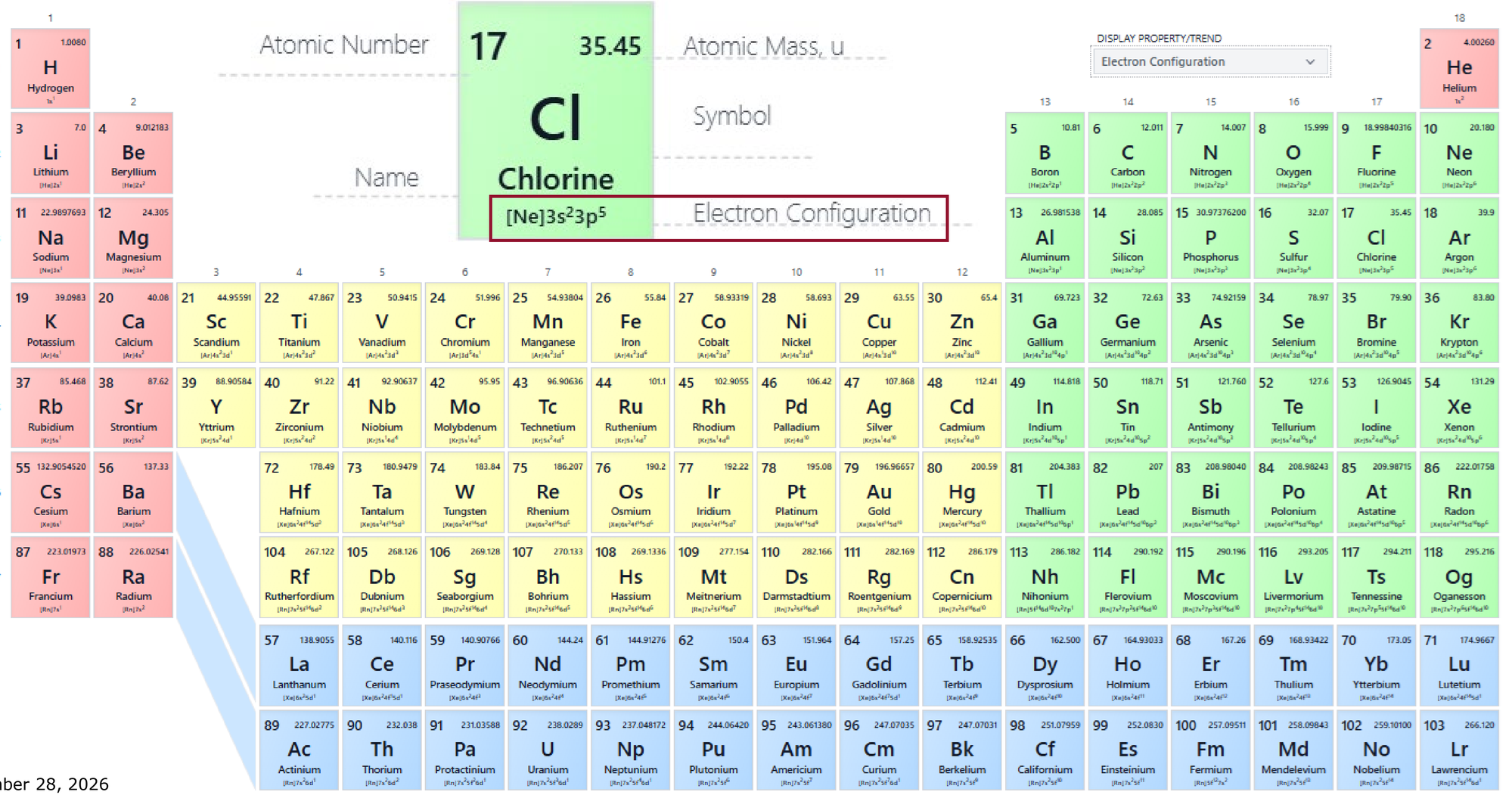
- All elements in a period have the same number of electron shells (principal quantum number n, from K to N) = **highest n used**
- The filling of each shell corresponds to a period
- Elements in the same period in f-block or p-block do not exhibit trends and similarities in properties
- In the d-block, trends across periods become significant, and in the f-block elements show a high degree of similarity across periods





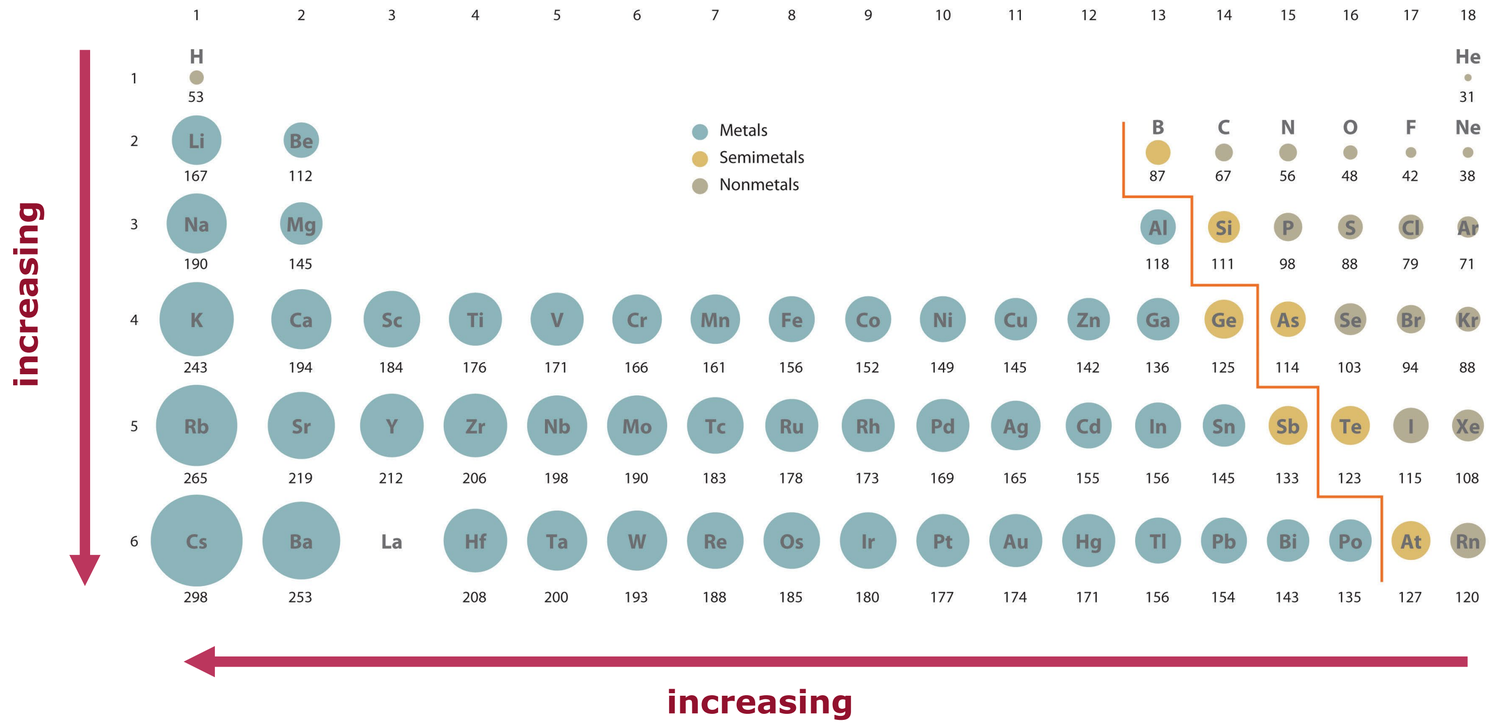


Periodic trend in electron configuration

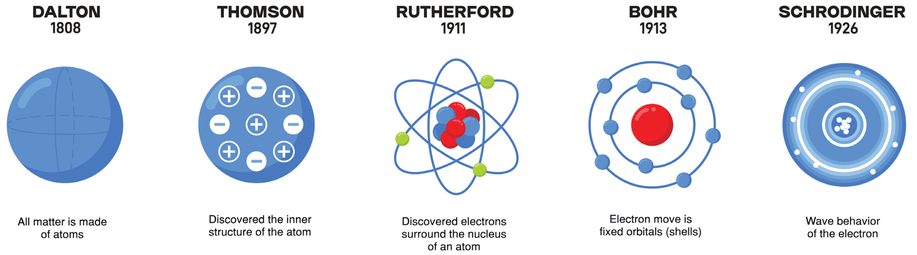


Periodic trend in atomic radius

- Atomic radii increase from top to bottom down a group (column)
  - Each step down adds a new electron shell (higher principal quantum number, n) → distance from the nucleus to outer electrons increases
- Atomic radii decrease from left to right across a period (row)
  - Number of protons (nuclear charge) increases, pulling electrons closer to the nucleus



Summary of Atomics models



After exploring how our understanding of the atom evolved, we see a clear pattern: each new model solved the problems of the previous one. **Dalton introduced the atom as indivisible, Thomson found electrons, Rutherford discovered the nucleus, Bohr explained quantized energy levels, and finally Schrödinger revealed the true quantum nature of electrons.**

But Schrödinger's model went beyond just "energy levels." He showed that electrons don't travel in fixed orbits like planets. Instead, they exist in regions of space where they are most likely to be found. These regions are called orbitals.

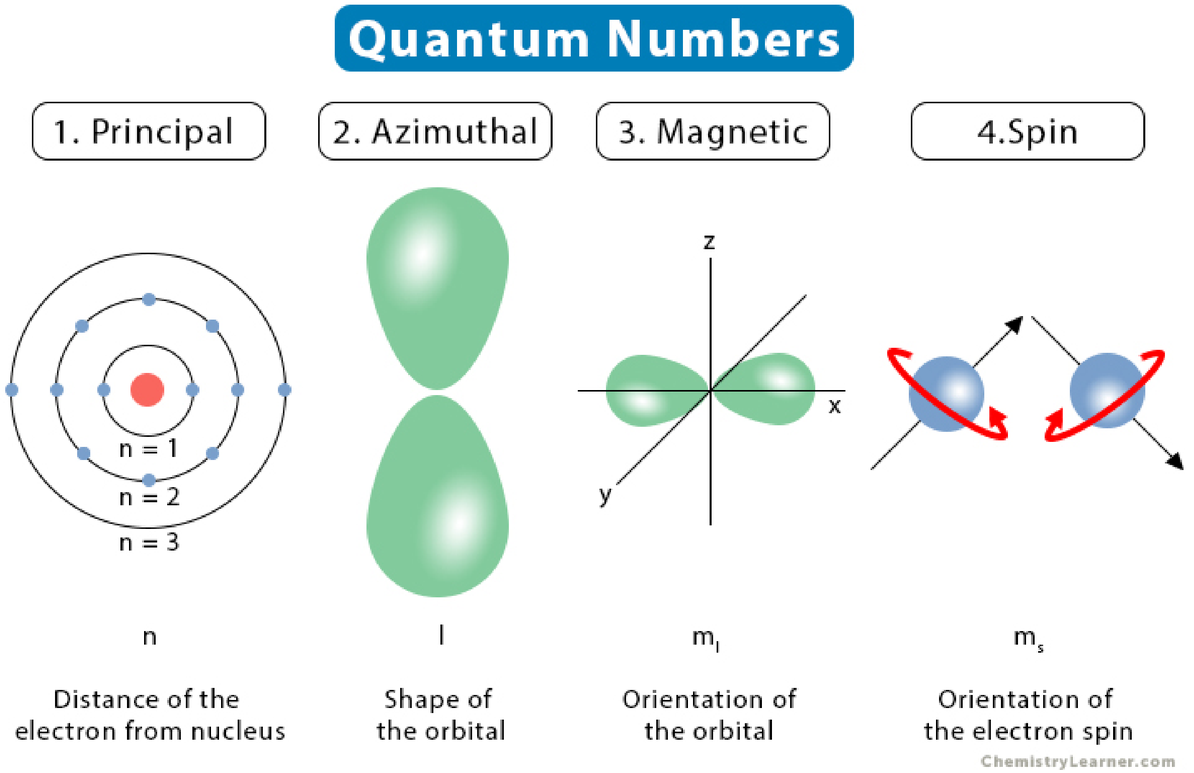
This was a major shift in thinking:

- From orbits (fixed circular paths in Bohr's model)
- To orbitals (3D probability clouds in Schrödinger's model).
- And orbitals come in different shapes (s, p, d, f), which explain why elements behave differently in chemical reactions and why the periodic table is structured the way it is.

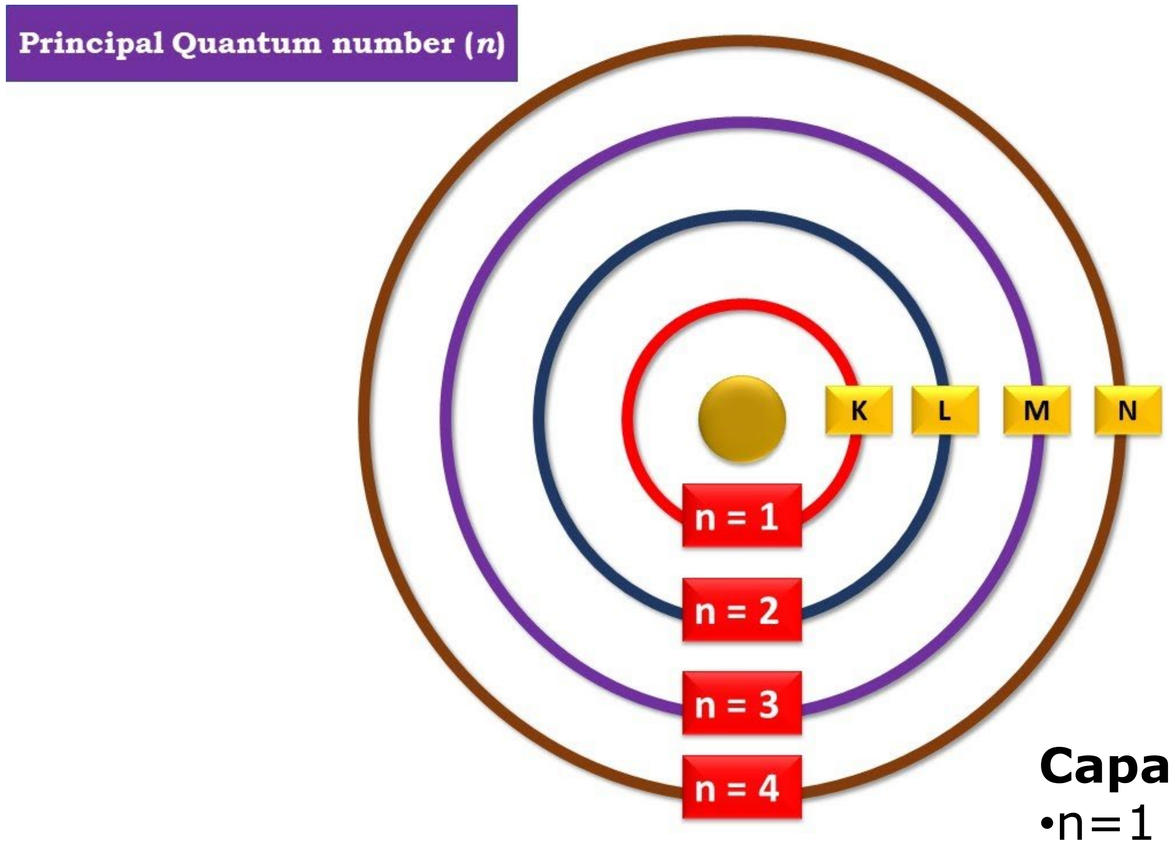
Orbital model

4 quantum numbers:

1. Principal quantum number (n): size of the orbital (**1**,**2**,**3**,...)
2. Angular quantum number (l): shape of the orbital (**s**, **p**, **d**, **f**)
3. Magnetic quantum number (m): orientation of the orbital in space (**0**, ±**1**, ±**2**,...)
4. Spin quantum number (s): intrinsic rotation of the electron (left or right: **+½ or -½**)



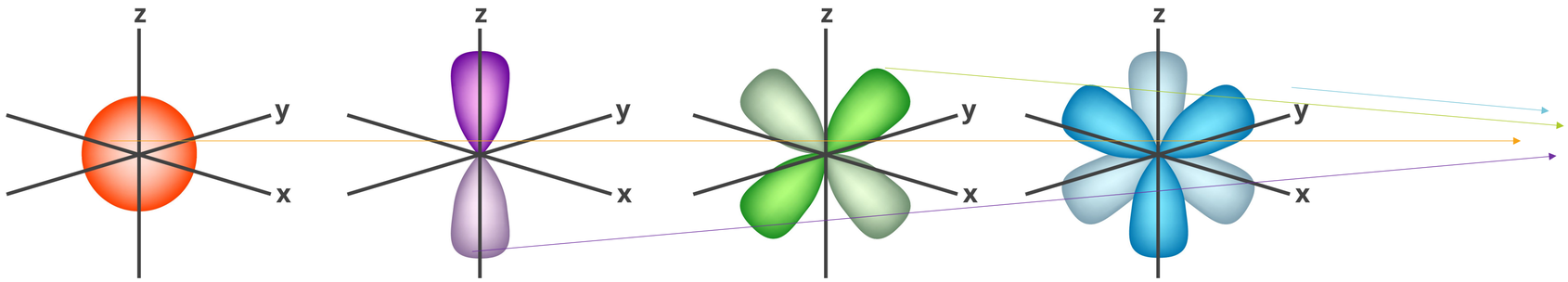
1. Principal quantum number (n): size of the orbital



Capacity per shell (first four):

- n=1 → 1s → 2 e^- (K)
- n=2 → 2s + 2p → 2 + 6 = 8 e^- (L)
- n=3 → 3s + 3p + 3d → 2 + 6 + 10 = 18 e^- (M)
- n=4 → 4s + 4p + 4d + 4f → 2 + 6 + 10 + 14 = 32 e^- (N)

2. Angular quantum number (l): shape of the orbital (n-1)



These regions are the probablity to find an electron

s orbital **p orbital** **d orbital** **f orbital**

l:0 l:1 l:2 l:3

Orbital type	Shape	Max electrons
s	spherical	2
p	dumbbell	6
d	cloverleaf	10
f	complex	14

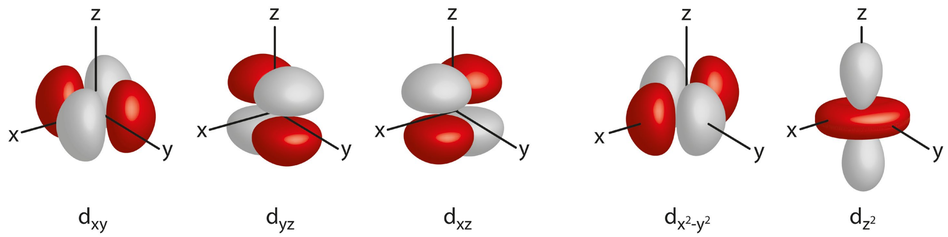
3. Magnetic quantum number (m): orientation of the orbital (-l…0…+l)

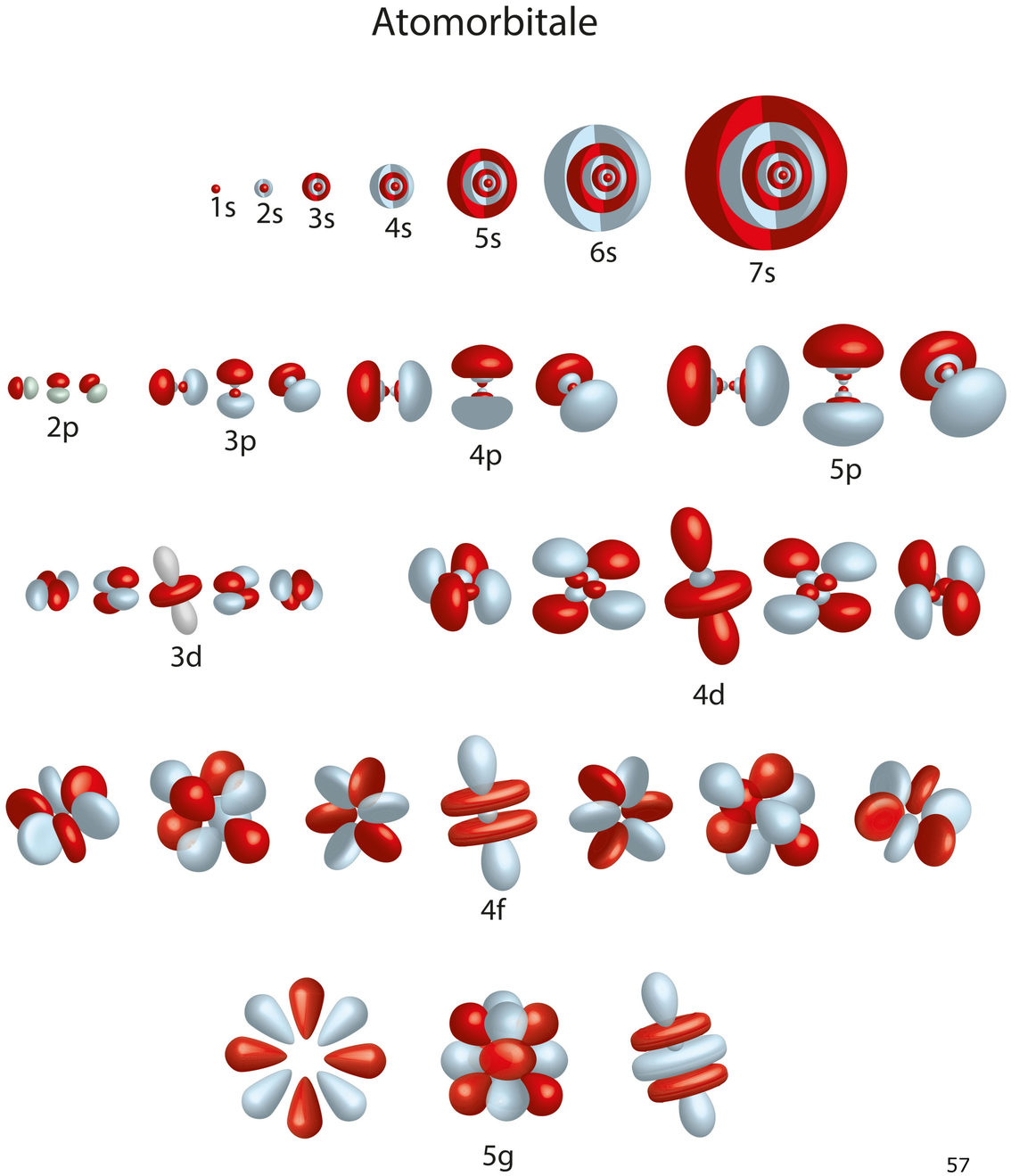
m = -l, -(l-1), -(l-2), 0, +(l-2), +(l-1), +l

Example d-Orbital:

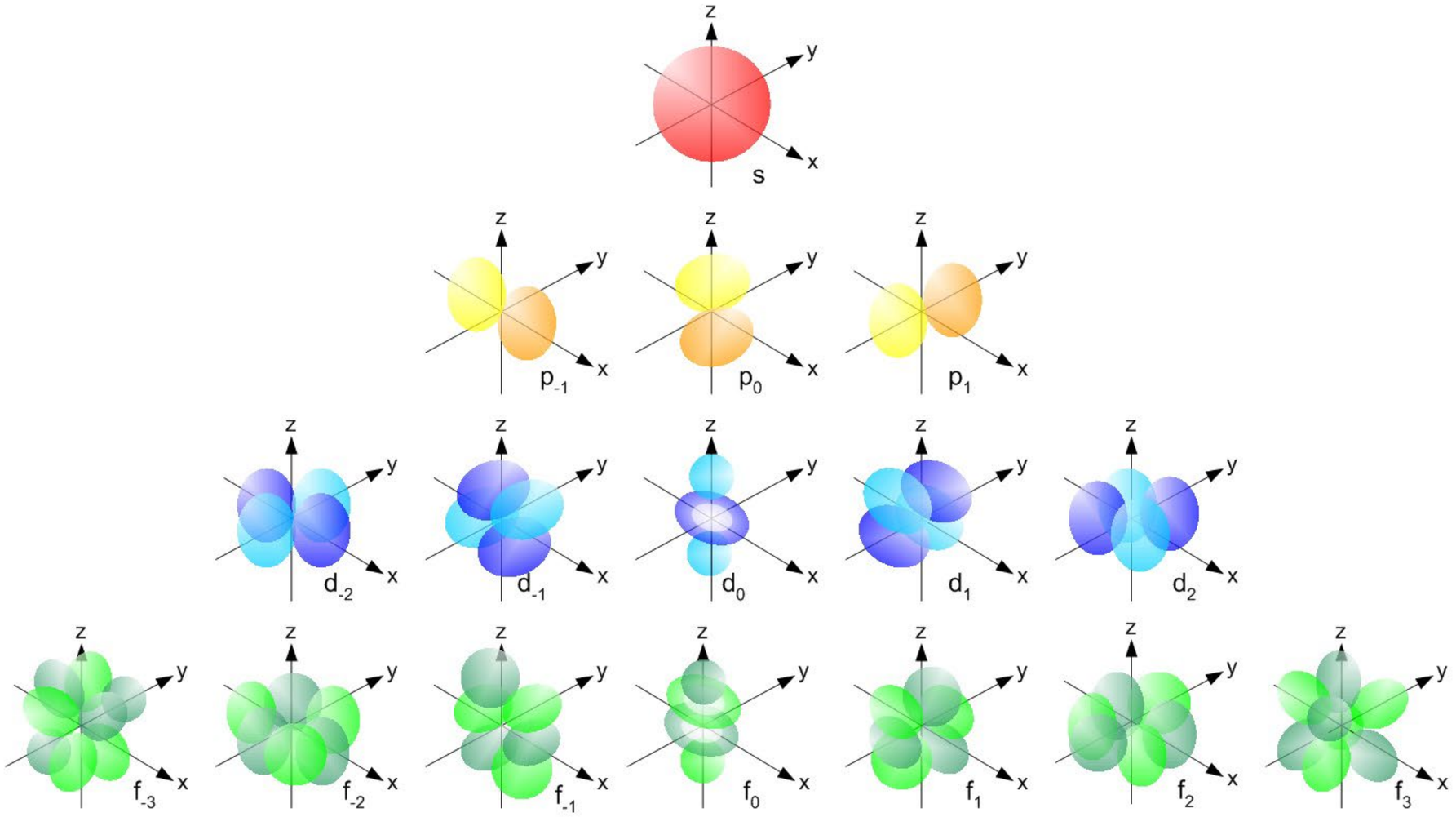
l = 2

m = -2, -1, 0, +1, +2



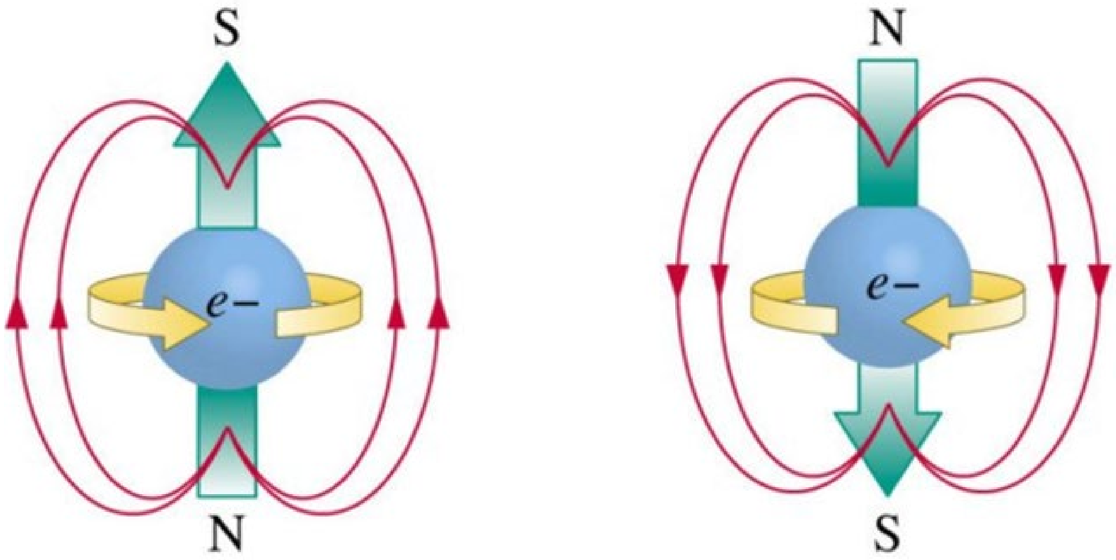


3. Magnetic quantum number (m): orientation of the orbital

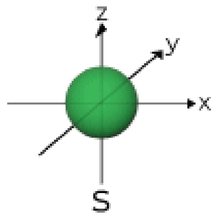
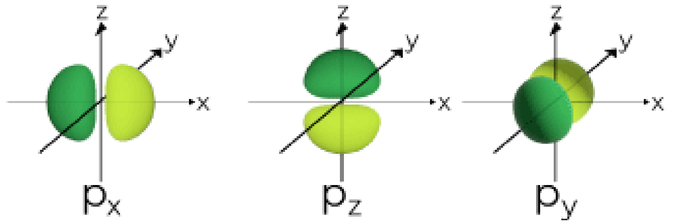
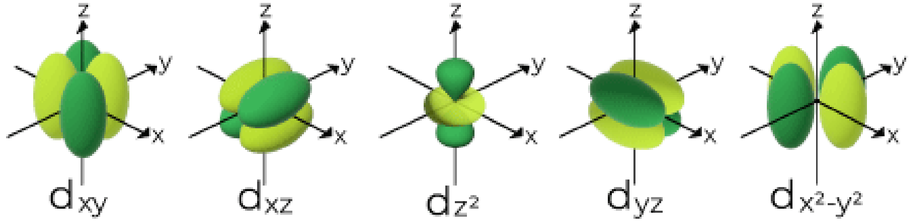
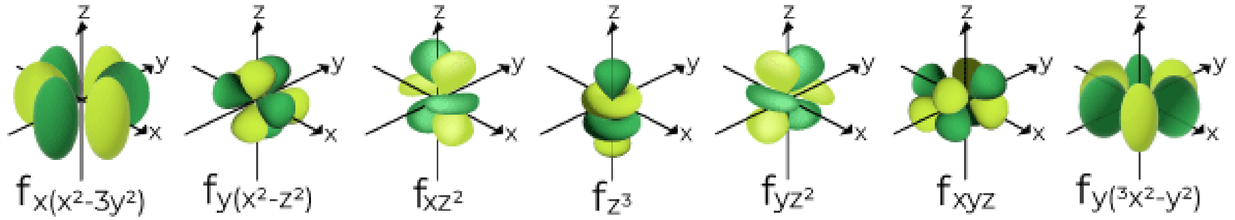


4. Spin quantum number (s): intrinsic rotation of the electron

To the left or to the right → +1/2 or -1/2 (**(+0.5 or -0.5)**



Quantum numbers and orbitals

s orbital		n = 1, 2, 3, … 7 l = 0 m = 0
p orbital		n = 2, 3, … 6 l = 1 m = 0, ±1
d orbital		n = 3, 4,and 5 l = 2 m = 0, ±1, ±2
f orbital		n = 4 l = 3 m = 0, ±1, ±2, ±3

Orbital model

Pauli exclusion principle:

2 electrons in an atom cannot have the same set of quantum numbers.

→ Therefore, a maximum of 2 electrons per orbital!

Number of electrons ($2n^2$)

Principal Q-number n	Shell	Azimuthal Q-number l	Subshell	Magnetic Quantum Number m	Spin Quantum Number s	Number of Electrons Accommodated
1	K	0	s	0	+½, −½	2
2	L	0	s	0	+½, −½	2
		1	p	+1, 0, −1	+½, −½	6
						8 (total)
3	M	0	s	0	+½, −½	2
		1	p	+1, 0, −1	+½, −½	6
		2	d	+2, +1, 0, −1, −2	+½, −½	10
						18 (total)
4	N	0	s	0	+½, −½	2
		1	p	+1, 0, −1	+½, −½	6
		2	d	+2, +1, 0, −1, −2	+½, −½	10
		3	f	+3, +2, +1, 0, −1, −2, −3	+½, −½	14
						32 (total)

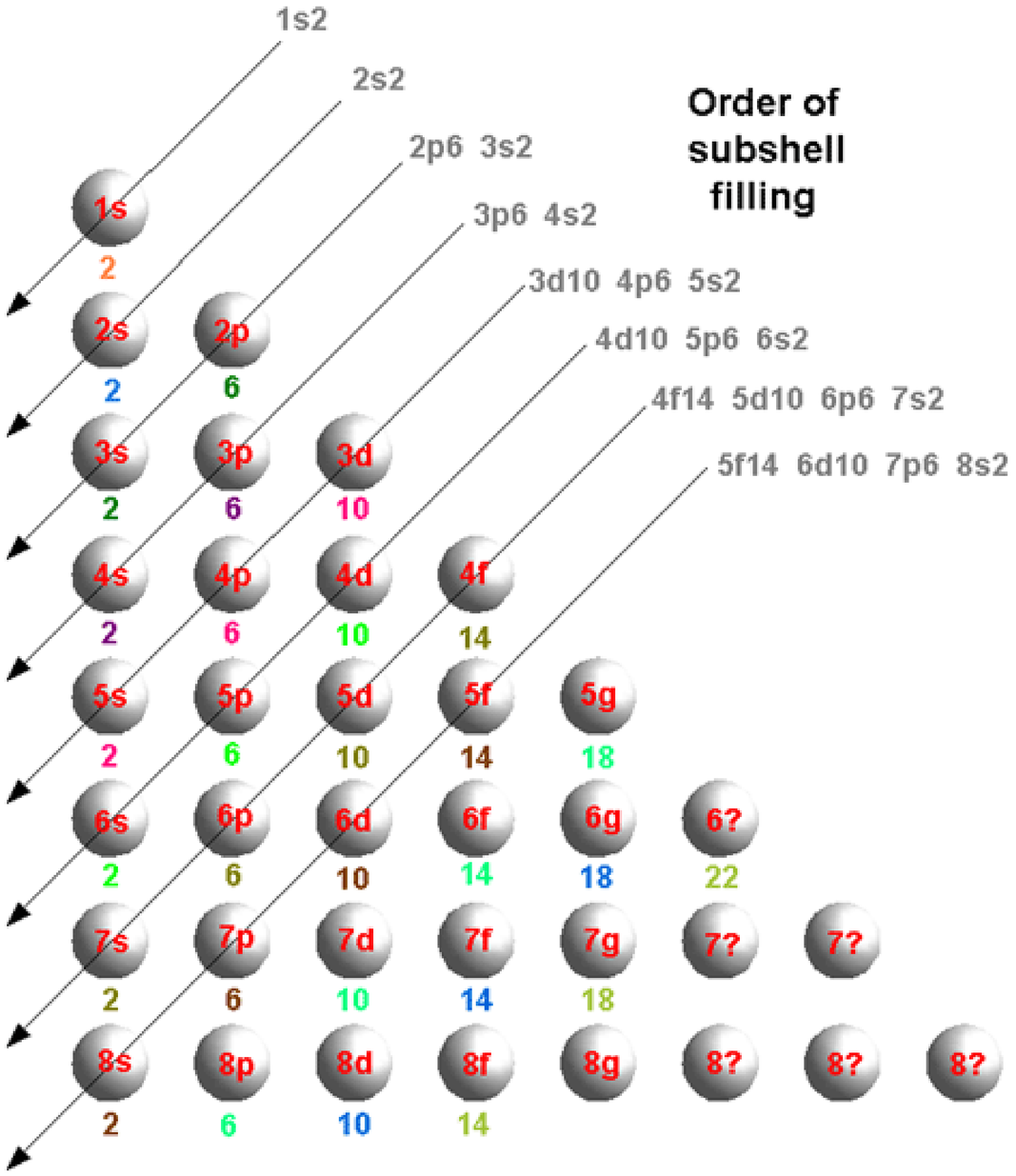
Filling orbitals: electron configuration

The orbitals are filled from the nearest s-orbital outward, aiming for the lowest energy state.

Main rule: outermost shell never has more than 8 electrons.

The outermost shell never has more than **8 electrons**.
- **1s** with 2 electrons
- **2s** with 2 electrons
- **2p** with 6 electrons
- **3s** with 2 electrons
- **3p** with 6 electrons

Example: Na: $1s^2 2s^2 2p^6 3s^1$ (11 electrons)



Filling orbitals: electron configuration

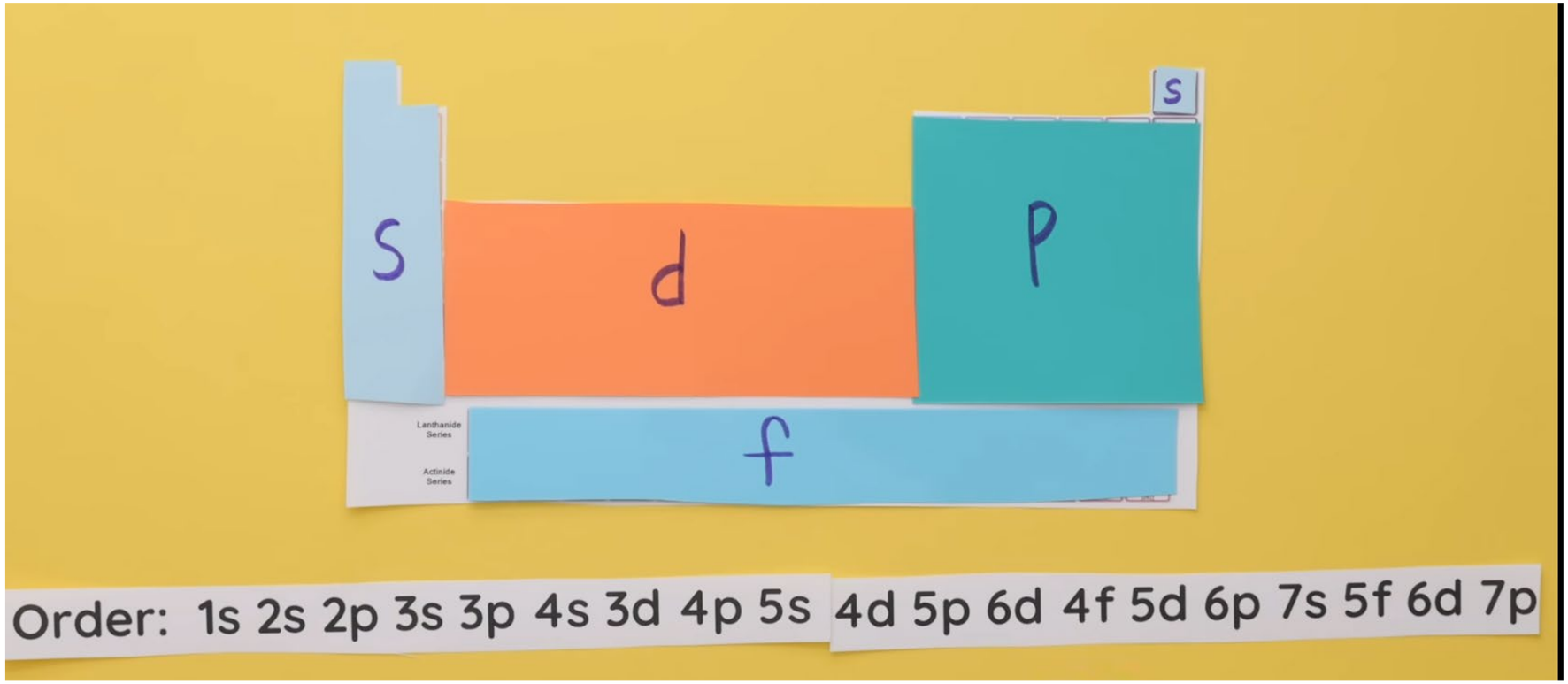
How to write a configuration:

Find the element’s **atomic number** (total electrons in the neutral atom).

Fill the subshells following the order above until you place all electrons.

Use superscripts to show how many electrons are in each subshell, e.g. **$2p^6$**.

Noble-gas shorthand: replace the inner, filled part with the previous noble gas in brackets.


s
s
d
p
Lanthanide Series
Actinide Series
f
Order: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6d 4f 5d 6p 7s 5f 6d 7p

Two must-know rules

Pauli Exclusion Principle

No two electrons in the same atom can have all four quantum numbers identical.

Practical effect: **Each orbital holds at most 2 electrons** and, when it holds 2, their **spins are opposite** (one +½, one −½).

Hund's Rule

Electrons prefer to occupy different orbitals of the same energy with parallel spins before pairing up.

"Bus seat rule": when you and your friends get on an empty bus, you spread out to empty seats (orbitals) before sharing a seat (pairing spins).

In a **p** subshell (3 orbitals), electrons go ↑ ↑ ↑ one per box before any orbital gets ↑↓.

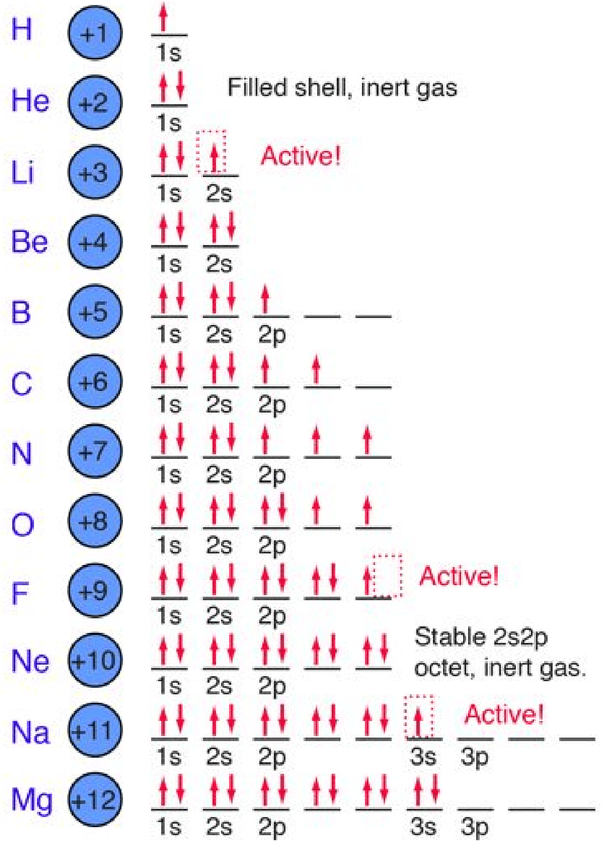
Hund’s rule / Valence electrons

p & d orbitals are first singly occupied (spins in the same direction) and only then spin pairing begins.

Elements where only one electron is missing to fill all orbitals are very reactive.

The outer electrons, those in unfilled shells, are called valence electrons.

Atomic Number	Element	Partial Orbital Diagram (4s, 3d, and 4p Sublevels Only) 4s	3d	4p	Full Electron **Configuration**	Condensed Electron **Configuration**
27	Co	↑↓	↑↓ ↑↓ ↑ ↑ ↑		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$	[Ar] 4 $s^2 3d^7$
28	Ni	↑↓	↑↓ ↑↓ ↑↓ ↑ ↑		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$	[Ar] 4 $s^2 3d^8$
29	Cu	↑	↑↓ ↑↓ ↑↓ ↑↓ ↑↓		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$	[Ar] $4s^1 3d^{10}$
30	Zn	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓		$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$	[Ar] $4s^2 3d^{10}$
31	Ga	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^1$	[Ar] 4 $s^2 3d^{10} 4p^1$
32	Ge	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑ ↑	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$	[Ar] 4 $s^2 3d^{10} 4p^2$
33	As	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑ ↑ ↑	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^3$	[Ar] 4 $s^2 3d^{10} 4p^3$
34	Se	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑↓ ↑ ↑	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^4$	[Ar] 4 $s^2 3d^{10} 4p^4$
35	Br	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$	[Ar] 4 $s^2 3d^{10} 4p^5$
36	Kr	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑↓	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$	[Ar] 4 $s^2 3d^{10} 4p^6$



Exercises

H (1):

He (2):

Li (3):

C (6):

N (7):

O (8):

Ne (10):

Na (11):

Exercises – Results

H (1): $1s^1$

He (2): $1s^2$

Li (3): $1s^2\ 2s^1$ = [He] $2s^1$

C (6): $1s^2\ 2s^2\ 2p^2$ = [He] $2s^2\ 2p^2$

N (7): $1s^2\ 2s^2\ 2p^3$ = [He] $2s^2\ 2p^3$

O (8): $1s^2\ 2s^2\ 2p^4$ = [He] $2s^2\ 2p^4$

Ne (10): $1s^2\ 2s^2\ 2p^6$ = [He] $2s^2\ 2p^6$

Na (11): $1s^2\ 2s^2\ 2p^6\ 3s^1$ = [Ne] $3s^1$

Exercises

Fill in the quantum numbers (n, l, m, s) and orbital types for Carbon (Z = 6).

Shell	n	l	Orbital Type	m	s

Shell	n	l	Orbital Type	m	s
K	1	0	s	0	+0.5 / -0.5
L	2	0	s	0	+0.5 / -0.5
L	2	1	p	-1	+0.5 / -0.5
L	2	1	p	0	+0.5 / -0.5

Exercises

Carbon (C)

Atomic number 6 → 6 e^-

Configuration: $1s^2\ 2s^2\ 2p^2$ = [He] $2s^2\ 2p^2$

Valence e^-: 4 (the $2s^2\ 2p^2$)

Hund & Pauli: in 2p, electrons spread out: 2p: [↑] [↑] [ ]

Typical bonding: four σ bonds (e.g., CH_4) → **sp^3**; with a double bond (as in C=C), each C is **sp^2**; with a triple (C≡C), each C is **sp**.

Example B: Nitrogen (N)

Z=7 → 7 e^-

Config: $1s^2\ 2s^2\ 2p^3$ = [He] $2s^2\ 2p^3$

Valence e^-: 5

2p diagram: [↑] [↑] [↑]

In **NH_3**, N forms 3 σ bonds + 1 lone pair → SN=4 → **sp^3** (trigonal pyramidal)

Exercises

Oxygen (O)

Atomic number:

Config:

Valence e^-: 6

2p diagram:

Exercises

Oxygen (O)

Z=8 → 8 e^-

Config: $1s^2\ 2s^2\ 2p^4$ = [He] $2s^2\ 2p^4$

Valence e^-: 6

2p diagram: [↑↓] [↑] [↑]

In **H_2O**, O forms 2 σ bonds + 2 lone pairs → SN=4 → **sp^3** (bent)

Tips

Counting π bonds as extra "regions": Don't. For SN, a double or triple bond counts as **one** σ bond region.

Forgetting lone pairs when counting SN: They matter for hybridization and shape.

Mixing "electron geometry" with "molecular shape": Electron geometry counts lone pairs; molecular shape focuses on atoms only (e.g., NH_3 is trigonal pyramidal even though electron geometry is tetrahedral).

Hund's rule violations: Don't pair electrons in p orbitals until each p orbital has one.

Pauli: Never put two same-spin arrows in the same box.

Group vs. period: Group (column) ≈ valence count (for main group). Period (row) = highest n used.

Memory helpers:

Filling order (say it aloud): 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s…

SN→hybrid: 2→sp, 3→sp^2, 4→sp^3.

Double bond = 1σ + 1π. **Triple** = 1σ + 2π.

More exercises self study

1.Na (11) → configuration? valence e^-?

2.Mg (12) → configuration? valence e^-?

3.C (6) → configuration? valence e^-?

4.N (7) → configuration? valence e^-?

5.O (8) → configuration? valence e^-?

Orbital diagrams (Hund + Pauli)

6) Draw the 2p orbital diagram for **N** and **O**.

Hybridization & shape

7) **CH_4**: hybridization at C? geometry? approx. angle?

8) **NH_3**: hybridization at N? geometry? approx. angle?

9) **H_2O**: hybridization at O? geometry? approx. angle?

10) **C_2H_4 ($H_2C{=}CH_2$)**: hybridization at each C? planar or not?

11) **C_2H_2 ($HC{\equiv}CH$)**: hybridization at each C? linear or not?

12) **CO_2 (O=C=O)**: hybridization at C? geometry at C?

13) **H_3O^+ (hydronium)**: hybridization at O? shape? (Hint: O has 3 σ to H and 1 lone pair.)

More exercises on Ilias...

More exercises self study- Answers

1. **Na**: $1s^2\ 2s^2\ 2p^6\ 3s^1$ = [Ne] $3s^1$; valence = 1
2. **Mg**: $1s^2\ 2s^2\ 2p^6\ 3s^2$ = [Ne] $3s^2$; valence = 2
3. **C**: $1s^2\ 2s^2\ 2p^2$ = [He] $2s^2\ 2p^2$; valence = 4
4. **N**: $1s^2\ 2s^2\ 2p^3$ = [He] $2s^2\ 2p^3$; valence = 5
5. **O**: $1s^2\ 2s^2\ 2p^4$ = [He] $2s^2\ 2p^4$; valence = 6
6. **N 2p**: [↑] [↑] [↑]; **O 2p**: [↑↓] [↑] [↑]
7. **CH_4**: C is **sp^3**; tetrahedral; ~109.5°
8. **NH_3**: N is **sp^3**; trigonal pyramidal; ~107°
9. **H_2O**: O is **sp^3**; bent; ~104.5°
10. **C_2H_4**: each C is **sp^2**; planar at each C (trigonal planar around C); ~120°
11. **C_2H_2**: each C is **sp**; linear; 180°
12. **CO_2**: C is **sp**; linear at C; 180°
13. **H_3O^+**: O is **sp^3** (3 σ + 1 lone pair); trigonal pyramidal